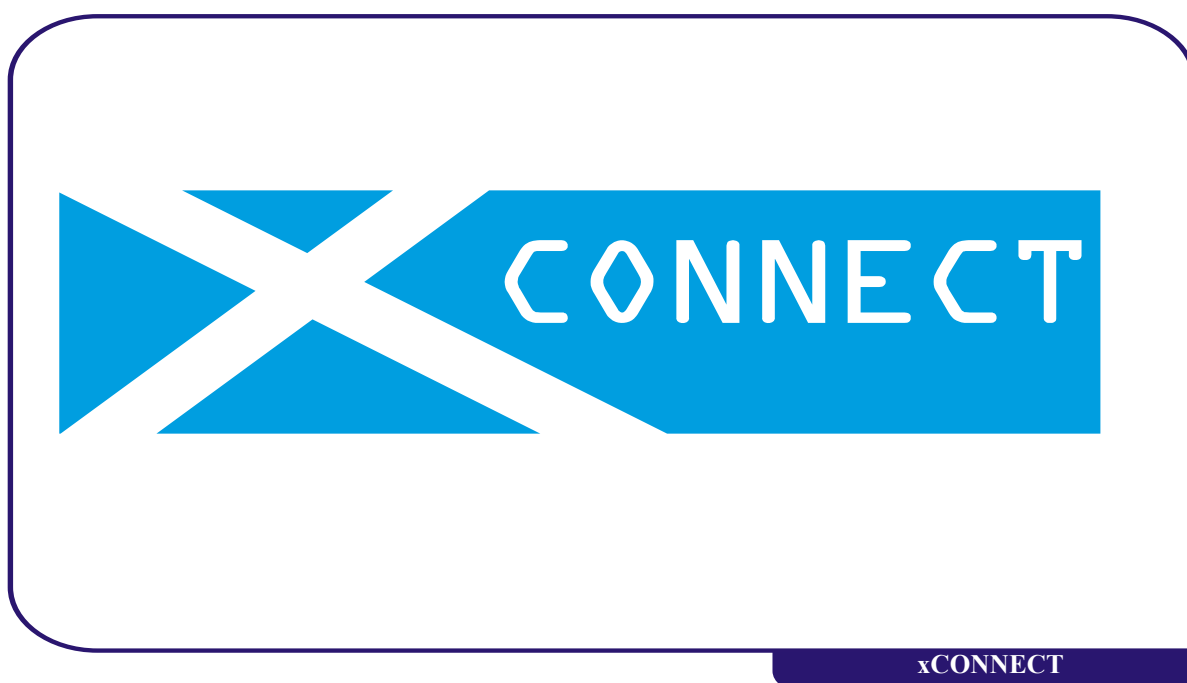




CONNECTIVITY



SUPERVISION SYSTEMS FOR CONTROLLERS:

- IC121C
- IC121CX
- IC281L
- IC208CX
- IC208CX (DRY)
- IC208CX (FC4ALL, FC4TAE)
- IC121CX (AQUAFREE)
- XR60CX
- tDRY
- iDRY

INTRODUCTION

xCONNECT is the name assigned by MTA to denote the set of technological solutions utilised for modularity and local or remote supervision of its units.

This manual describes the supervision of units with controller IC121C, IC121CX, IC281L, IC208CX, XR60CX, tDRY, iDRY.

Supervision makes it possible to access operating data of a unit and adjust its parameters by means of a MODBUS protocol.

The modularity system makes it possible to connect several chiller/heat pumps together in such a way as to obtain higher overall cooling capacity in order to assure superior precision of temperature control.

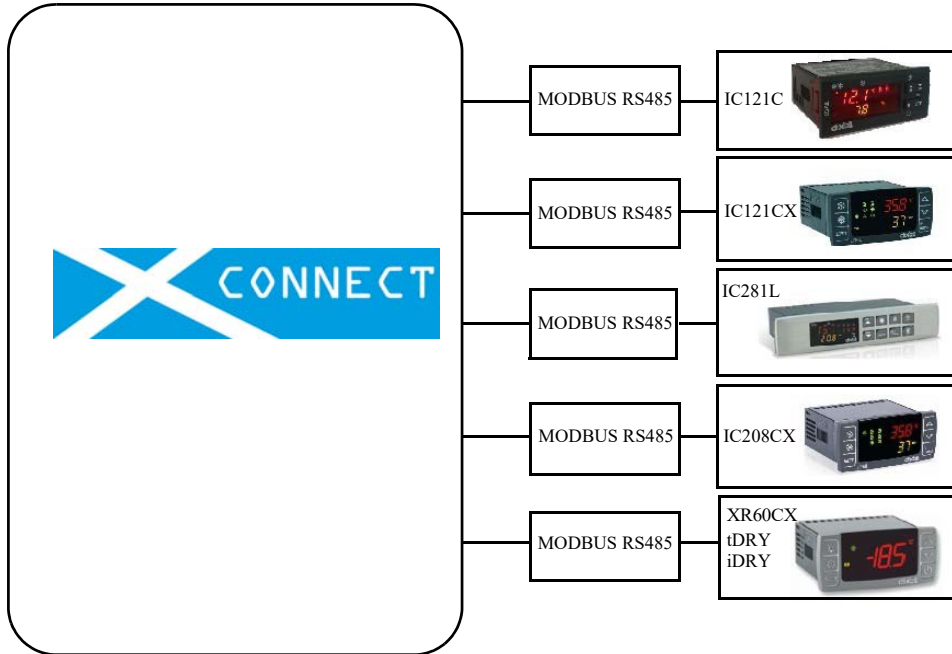


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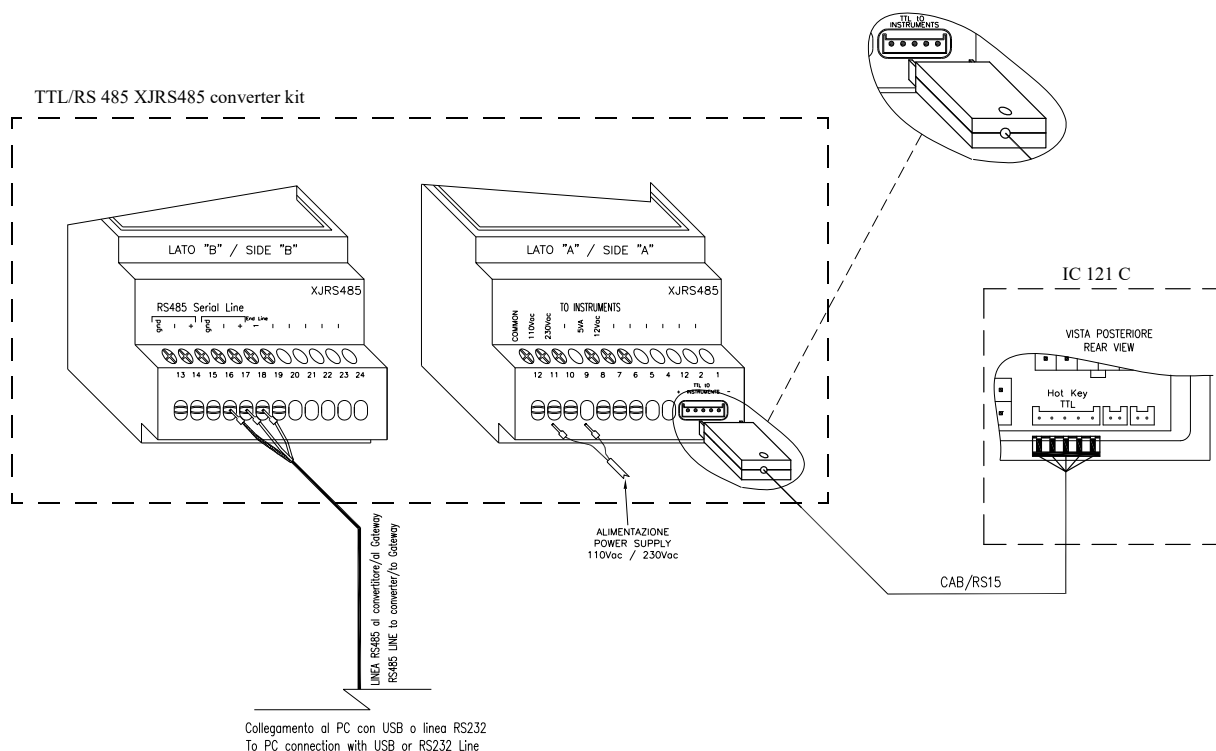
H/CG - H/OC - EOLO - H/TA - MCTA - H/TAT - MCTAT - TAEevo - TWEEvo - HAEevo - H/ON



1. MODBUS-RTU

1.1 Connection drawing

The drawing below shows how to connect the unit to a Modbus network:



1.2 Configuration

Communication is configured as follows:

Baud Rate = 9600 bps

Data Length = 8 bit

Parity = None

Stop Bit = 1

1.3 How to set the device address

Press **SET** +  buttons for 3 sec. to enter the programming procedure, then press **SET** until reaching **CF34** parameter.

Use  and  buttons to modify the value.

Press **SET** again to store the new value.

1.4 Variables

IDENTIFICATION AND CONFIGURATION (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
1	Product family code/Firmware version	1	Bit15-Bit8: product family code; Bit7-Bit0: Firmware Version
6	Firmware version date	1	Bit15-Bit11:Day; Bit10-Bit7:Month; Bit6-Bit0:Year

PROBE STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
257	1st probe value	1	
259	2nd probe value	1	
261	3rd probe value	1	
263	4th probe value	1	

NOTE

Consult to the electrical diagram of the machine to see that the probe is related to the "value of probe 1." PB1= Probe value 1 / PB2= Probe value 2 /

DIGITAL INPUT STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
516	Digital input status (4th)	1	Bit0 Energy saving Bit1 Remote heat pump/chiller (only enabled with param. CF028=1) Bit2 Low pressure switch Bit3 High pressure switch Bit4 Condensing units on Bit5 Condensing unit chiller on Bit6 Condensing unit heat pump on Bit7 Antifreeze Bit8 1st compressor thermal overload Bit9 Condensing fan thermal overload Bit10 Delivery fan thermal overload/Flow switch Bit11 Remote off Bit12 Remote chiller (only enabled with param. CF028=1) Bit13 2nd compressor thermal overload Bit14 Condensing units 2nd step request Bit15 Defrost end
517	Digital input status (5th)	1	Bit0 Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Bit7 Bit8 Unspecified alarm from digital input on Bit9 Bit10 Bit11 Bit12 Bit13 Bit14 Bit15

USER PARAMETERS (writing)			
MODBUS ADDRESS (DEC)	PARAMETER	DISPLAY REF.	SIZE (Word 16bit)
769	Summer setpoint (during writing the display must show the main mask)	St01	1
771	Winter setpoint (during writing the display must show the main mask)	St03	1

UNIT FUNCTIONAL STATUS (read-write)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
1281	Unit status (1st)	1	Bit0 Unit on-off (Send Modbus command with only Bit0 on 1 to switch the unit off - during writing the display must show the main mask) Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Bit7 Bit8 Unit on/off0: stand-by or remote-off 1: chiller or heat pump on (read only) Bit9 Bit10 Bit11 Bit12 Bit13 Bit14 Bit15
1283	Unit status (3rd)	1	Bit0 Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Chiller mode switch-on enabling (Send Modbus command with only Bit6 and Bit14 on 1 to switch the unit on in chiller mode - during writing the display must show the main mask) Bit7 Heat pump mode switch-on enabling (Send Modbus command with only Bit7 and Bit15 on 1 to switch the unit on in heat pump mode - during writing the display must show the main mask) Bit8 Bit9 Bit10 Unit on/off from digital input (read only) Bit11 Bit12 Bit13 Bit14 Chiller mode switch-on (Send Modbus command with only Bit6 and Bit14 on 1 to switch the unit on in chiller mode - during writing the display must show the main mask) Bit15 Heat pump mode switch-on (Send Modbus command with only Bit7 and Bit15 on 1 to switch the unit on in heat pump mode - during writing the display must show the main mask)

FUNCTION PARAMETERS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
1537	Real setpoint (if enabled)	1	
1538	Chiller setpoint	1	
1539	Heat pump setpoint	1	

DIGITAL OUTPUT STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
2049	Relay output status (1st)	1	Bit0 Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Bit7 Bit8 Bit9 Bit10 Bit11 Open collector/alarm relay Bit12 Bit13 Bit14 Bit15
2050	Relay output status (2nd)	1	Bit0 Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Bit7 Bit8 Compressor 1 relay Bit9 Compressor 2 relay Bit10 Delivery fan relay Bit11 Condensing fan relay Bit12 Anti-freeze relay Bit13 Pump relay Bit14 Solenoid valve relay on water side Bit15

ANALOGUE OUTPUT STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
2305	Analogue output 1 value	1	
2307	Analogue output 2 value	1	

ALARM STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
3332	Functional alarms (4th)	1	Bit0 A04 - Pb2 probe low temperature Bit1 A05 - High condensing temperature/ pressure Bit2 A06 - Low condensing temperature/ pressure Bit3 A07 - Antifreeze Bit4 Bit5 A08 - Flow switch/water differential pressure switch/delivery fan thermal overload Bit6 A09 - Compressor 1 thermal overload Bit7 A10 - Compressor 2 thermal overload Bit8 EE - Eeprom Bit9 P1 - Pb1 probe damaged Bit10 P2 - Pb2 probe damaged Bit11 P3 - Pb3 probe damaged Bit12 P4 - Pb4 probe damaged Bit13 A01 - High pressure switch Bit14 A02 - Low pressure switch Bit15 A03 - Pb1 probe low temperature

3333	Functional alarms (5th)	1	Bit0 Bit1 Bit2 Bit3 Bit4 Bit5 ALOC - Unit lockout Bit6 A20 - Pb2 low temperature Bit7 A21 - Pb2 high temperature Bit8 A11 - Condensing fan thermal overload Bit9 A12 - Defrost end Bit10 A13 - Compressor 1 maintenance Bit11 A14 - Compressor 2 maintenance Bit12 A15 - Pump / delivery fan maintenance Bit13 Bit14 Bit15 Afr - Network frequency
3334	Functional alarms (6th)	1	

LOAD HOURS (read-write)

MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
3585	Compressor 1 operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	
3586	Compressor 2 operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	
3589	Pump operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	

ALARMS TRIPPED PER HOUR (reading)

MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
3841	Low pressure switch (AO02)	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3842	Pb2 probe low temperature AO04 - always 0 in non air-air units)	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3843	Low condensing temperature/pressure (A06)	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3844	Flow switch / delivery fan thermal overload (A08)	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3845	Anti-freeze (AO04 - always 0 in air-air units)	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3846	Compressor 1 thermal overload (A09)	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3847	Compressor 2 thermal overload (A10)	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3848	Unspecified machine lockout (ALOC)	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold

IC121CX AQUAFREE



2. MODBUS-RTU

2.1 Connection drawing

For the connection diagram, refer to the wiring diagram of the Aquafree modules.

2.2 Configuration

Communication is configured as follows:

Baud Rate = 9600 bps

Data Length = 8 bit

Parity = None

Stop Bit = 1

2.3 How to set the device address

Press **SET** +  buttons for 3 sec. to enter the programming procedure, then press **SET** until reaching **CF22** parameter.

Use  and  buttons to modify the value.

Press **SET** again to store the new value.

2.4 Variables

PROBE STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
277	Temperature probe at water inlet	1	
278	Ambient temperature probe	1	
279	Temperature probe at water outlet	1	
280	Status of temperature probe at water inlet	1	
281	Status of ambient temperature probe	1	
282	Status of temperature probe at water outlet	1	

DIGITAL VARIABLES (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE Bit	NOTE
533	Communication blink	1	
534	Fan overload alarm	1	
535	Freecooling probe alarm	1	
536	Ambient probe alarm	1	
537	High temperature alarm for freecooling probe	1	
538	Low temperature alarm for freecooling probe	1	
539	Pump alarm	1	
540	Pressure switch alarm	1	
541	Alarm for water outlet probe	1	
542	High temperature alarm for water outlet probe	1	
543	Low temperature alarm for water outlet probe	1	
544	Available alarm	1	

545	Pump relay status	1	
546	Alarm relay status	1	

DIGITAL VARIABLES (reading-writing)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE Bit	NOTE
518	On/Off	1	

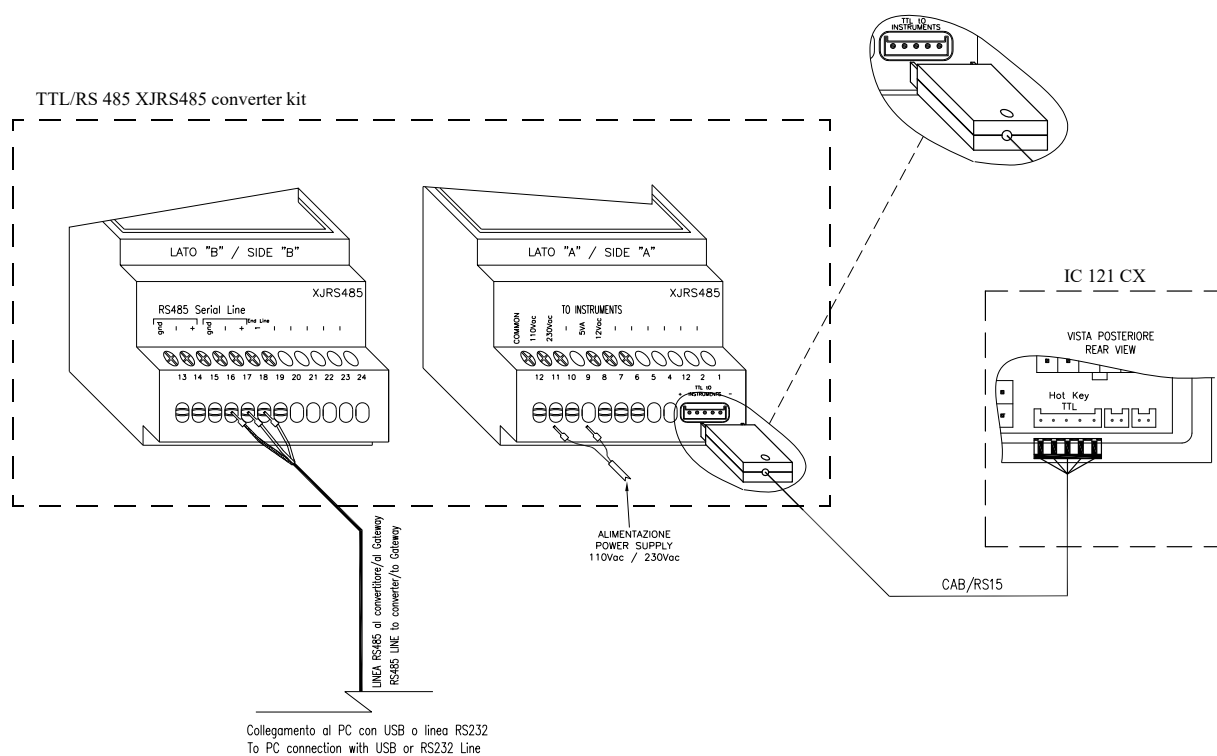
IC121CX T3L - CY - HCY - OC



3. MODBUS-RTU

3.1 Connection drawing

The drawing below shows how to connect the unit to a Modbus network:



3.2 Configuration

Communication is configured as follows:

Baud Rate = 9600 bps

Data Length = 8 bit

Parity = None

Stop Bit = 1

3.3 How to set the device address

Press **SET** + buttons for 3 sec. to enter the programming procedure, then press **SET** until reaching CF44 parameter.

Use and buttons to modify the value.

Press **SET** again to store the new value.

3.4 Variables

IDENTIFICATION AND CONFIGURATION (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
1	Product family code/Firmware version	1	Bit15-Bit8: product family code; Bit7-Bit0: Firmware Version
6	Firmware version date	1	Bit15-Bit11:Day; Bit10-Bit7:Month; Bit6-Bit0:Year

PROBE STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
257	1st probe value	1	
259	2nd probe value	1	
261	3rd probe value	1	
263	4th probe value	1	

NOTE

Consult to the electrical diagram of the machine to see that the probe is related to the "value of probe 1." $PB1 = \text{Probe value 1} / PB2 = \text{Probe value 2} / \dots$

DIGITAL INPUT STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
516	Digital input status (4th)	1	Bit0 Energy saving Bit1 Remote heat pump/chiller (only enabled with param. CF028=1) Bit2 Low pressure switch Bit3 High pressure switch Bit4 Condensing units on Bit5 Condensing unit chiller on Bit6 Condensing unit heat pump on Bit7 Antifreeze Bit8 1st compressor thermal overload Bit9 Condensing fan thermal overload Bit10 Delivery fan thermal overload/Flow switch Bit11 Remote off Bit12 Remote chiller (only enabled with param. CF028=1) Bit13 2nd compressor thermal overload Bit14 Condensing units 2nd step request Bit15 Defrost end
517	Digital input status (5th)	1	Bit0 Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Bit7 Bit8 Unspecified alarm from digital input on Bit9 Bit10 Bit11 Bit12 Bit13 Bit14 Bit15

USER PARAMETERS (writing)			
MODBUS ADDRESS (DEC)	PARAMETER	DISPLAY REF.	SIZE (Word 16bit)
769	Summer setpoint (during writing the display must show the main mask)	St01	1
771	Winter setpoint (during writing the display must show the main mask)	St03	1

UNIT FUNCTIONAL STATUS (read-write)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
1281	Unit status (1st)	1	Bit0 Unit on-off (Send Modbus command with only Bit0 on 1 to switch the unit off - during writing the display must show the main mask) Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Bit7 Bit8 Unit on/off0: stand-by or remote-off 1: chiller or heat pump on (read only) Bit9 Bit10 Bit11 Bit12 Bit13 Bit14 Bit15
1283	Unit status (3rd)	1	Bit0 Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Chiller mode switch-on enabling (Send Modbus command with only Bit6 and Bit14 on 1 to switch the unit on in chiller mode - during writing the display must show the main mask) Bit7 Heat pump mode switch-on enabling (Send Modbus command with only Bit7 and Bit15 on 1 to switch the unit on in heat pump mode - during writing the display must show the main mask) Bit8 Bit9 Bit10 Unit on/off from digital input (read only) Bit11 Bit12 Bit13 Bit14 Chiller mode switch-on (Send Modbus command with only Bit6 and Bit14 on 1 to switch the unit on in chiller mode - during writing the display must show the main mask) Bit15 Heat pump mode switch-on (Send Modbus command with only Bit7 and Bit15 on 1 to switch the unit on in heat pump mode - during writing the display must show the main mask)

FUNCTION PARAMETERS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
1537	Real setpoint (if enabled)	1	
1538	Chiller setpoint	1	
1539	Heat pump setpoint	1	

DIGITAL OUTPUT STATUS (reading)

MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
2049	Relay output status (1st)	1	Bit0 Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Bit7 Bit8 Bit9 Bit10 Bit11 Open collector/alarm relay Bit12 Bit13 Bit14 Bit15
2050	Relay output status (2nd)	1	Bit0 Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Bit7 Bit8 Compressor 1 relay Bit9 Compressor 2 relay Bit10 Delivery fan relay Bit11 Condensing fan relay Bit12 Anti-freeze relay Bit13 Pump relay Bit14 Solenoid valve relay on water side Bit15

ANALOGUE OUTPUT STATUS (reading)

MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
2305	Analogue output 1 value	1	
2307	Analogue output 2 value	1	

ALARM STATUS (reading)

MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
3332	Functional alarms (4th)	1	Bit0 A04 - Pb2 probe low temperature Bit1 A05 - High condensing temperature/ pressure Bit2 A06 - Low condensing temperature/ pressure Bit3 A07 - Antifreeze Bit4 Bit5 A08 - Flow switch/water differential pressure switch/delivery fan thermal overload Bit6 A09 - Compressor 1 thermal overload Bit7 A10 - Compressor 2 thermal overload Bit8 EE - Eeprom Bit9 P1 - Pb1 probe damaged Bit10 P2 - Pb2 probe damaged Bit11 P3 - Pb3 probe damaged Bit12 P4 - Pb4 probe damaged Bit13 A01 - High pressure switch Bit14 A02 - Low pressure switch Bit15 A03 - Pb1 probe low temperature

3333	Functional alarms (5th)	1	Bit0 Bit1 Bit2 Bit3 Bit4 Bit5 ALOC - Unit lockout Bit6 A20 - Pb2 low temperature Bit7 A21 - Pb2 high temperature Bit8 A11 - Condensing fan thermal overload Bit9 A12 - Defrost end Bit10 A13 - Compressor 1 maintenance Bit11 A14 - Compressor 2 maintenance Bit12 A15 - Pump / delivery fan maintenance Bit13 Bit14 Bit15 Afr - Network frequency
3334	Functional alarms (6th)	1	

LOAD HOURS (read-write)

MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
3585	Compressor 1 operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	
3586	Compressor 2 operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	
3589	Pump operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	

ALARMS TRIPPED PER HOUR (reading)

MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
3841	Low pressure switch (AO02)	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3842	Pb2 probe low temperature AO04 - always 0 in non air-air units)	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3843	Low condensing temperature/pressure (A06)	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3844	Flow switch / delivery fan thermal overload (A08)	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3845	Anti-freeze (A07 - always 0 in air-air units)	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3846	Compressor 1 thermal overload (A09)	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3847	Compressor 2 thermal overload (A10)	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3848	Unspecified machine lockout (ALOC)	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold

IC281L

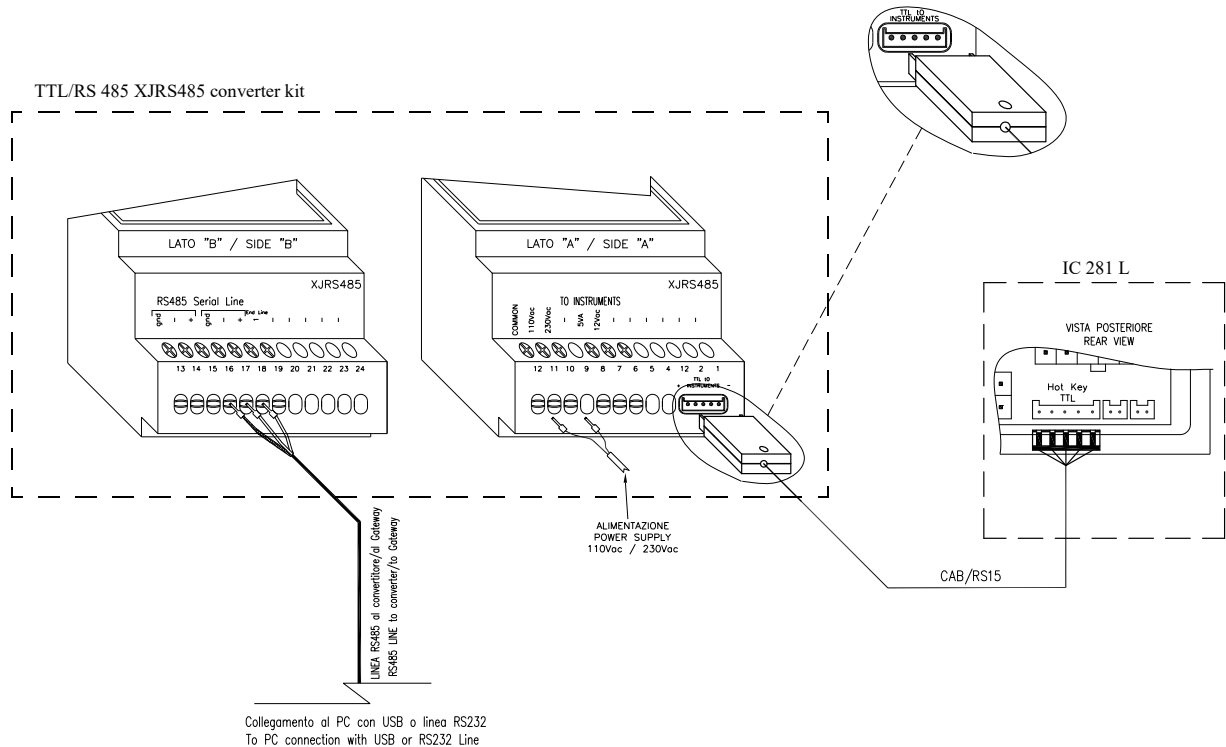
H/OC - NET - H/TAT - MCTAT - TWEEvo - TAEevo - H/ON - H/TA - MCTA



4. MODBUS-RTU

4.1 Connection drawing

The drawing below shows how to connect the unit to a Modbus network:



4.2 Configuration

Communication is configured as follows:

Baud Rate = 9600 bps

Data Length = 8 bit

Parity = None

Stop Bit = 1

4.3 How to set the device address

Press **SET** + buttons for 3 sec. to enter the programming procedure, then press **SET** until reaching **CF84** parameter.

Use and buttons to modify the value.

Press **SET** again to store the new value.

4.4 Variables

IDENTIFICATION AND CONFIGURATION (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
1	Product family code/Firmware version	1	Bit15-Bit8: product family code; Bit7-Bit0: Firmware Version
6	Firmware version date	1	Bit15-Bit11: Day Bit10-Bit7: Month; Bit6-Bit0: Year

PROBE STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
257	1st probe value	1	
259	2nd probe value	1	
261	3rd probe value	1	
263	4th probe value	1	
265	5th probe value	1	
267	6th probe value	1	
269	7th probe value	1	
271	8th probe value	1	
273	9th probe value	1	
275	10th probe value	1	
277	11th probe value	1	
279	12th probe value	1	

NOTE

Consult to the electrical diagram of the machine to see that the probe is related to the "value of probe 1." PB1 = Probe value 1 / PB2 = Probe value 2 /

DIGITAL INPUT STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
513	Digital input status (1st)	1	Bit0 Circuit 2 high pressure switch Bit1 Circuit 1 low pressure switch Bit2 Circuit 2 low pressure switch Bit3 Compressor 1 high pressure Bit4 Compressor 2 high pressure Bit5 Compressor 3 high pressure Bit6 Compressor 4 high pressure Bit7 Compressor 5 high pressure Bit8 Remote heat pump Bit9 Remote off Bit10 Remote chiller Bit11 Flow switch / delivery fan thermal overload Bit12 Hot side flow switch Bit13 Circuit 1 Anti-freeze Bit14 Circuit 2 Anti-freeze Bit15 Circuit 1 high pressure switch

514	Digital input status (2nd)	1	Bit0 Circuit 2 Condensing fan thermal overload Bit1 Condensing fan common thermal overload Bit2 Evaporator 1 water pump thermal overload Bit3 Backup evaporator water pump thermal overload Bit4 Condenser 1 water pump thermal overload Bit5 Backup condenser water pump thermal overload Bit6 Circuit 1 recovery request Bit7 Circuit 2 recovery request Bit8 Compressor 6 high pressure Bit9 Compressor 1 thermal overload Bit10 Compressor 2 thermal overload Bit11 Compressor 3 thermal overload Bit12 Compressor 4 thermal overload Bit13 Compressor 5 thermal overload Bit14 Compressor 6 thermal overload Bit15 Circuit 1 Condensing fan thermal overload
515	Digital input status (3rd)	1	Bit0 Compressor 6 oil pressure switch Bit1 Circuit 1 pump-down pressure switch Bit2 Circuit 2 pump-down pressure switch Bit3 General alarm input Bit4 Temperature control request (condensing unit) Bit5 Cold fluid request (condensing unit) Bit6 Hot fluid request (condensing unit) Bit7 Request for capacity control no. 1 compressor no. 1 (condensing unit) Bit8 Circuit 1 defrost end Bit9 Circuit 2 defrost end Bit10 Energy saving Bit11 Compressor 1 oil pressure switch Bit12 Compressor 2 oil pressure switch Bit13 Compressor 3 oil pressure switch Bit14 Compressor 4 oil pressure switch Bit15 Compressor 5 oil pressure switch
516	Digital input status (4th)	1	Bit0 Request for capacity control no. 2 compressor no. 3 (condensing unit) Bit1 Request for capacity control no. 3 compressor no. 3 (condensing unit) Bit2 Request for compressor no. 4 (condensing unit) Bit3 Request for capacity control no. 1 compressor no. 4 (condensing unit) Bit4 Request for capacity control no. 2 compressor no. 4 (condensing unit) Bit5 Request for capacity control no. 3 compressor no. 4 (condensing unit) Bit6 Request for compressor no. 5 (condensing unit) Bit7 Request for compressor no. 6 (condensing unit) Bit8 Request for capacity control compressor no. 1 (condensing unit) Bit9 Request for capacity control no. 3 compressor no. 1 (condensing unit) Bit10 Request for compressor no. 2 (condensing unit) Bit11 Request for capacity control no. 1 compressor no. 2 (condensing unit) Bit12 Request for capacity control no. 2 compressor no. 2 (condensing unit) Bit13 Request for capacity control no. 3 compressor no. 2 (condensing unit) Bit14 Request for compressor no. 3 (condensing unit) Bit15 Request for capacity control no. 1 compressor no. 3 (condensing unit)

USER PARAMETERS (read-write)			
MODBUS ADDRESS (DEC)	PARAMETER	DISPLAY REF.	SIZE (Word 16bit)
769	Summer setpoint (during writing the display must show the main mask)	St01	1
772	Winter setpoint (during writing the display must show the main mask)	St04	1

UNIT FUNCTIONAL STATUS (read-write)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
1281	Unit status (1st)	1	Bit0 Unit off (Send Modbus command with only Bit0 on 1 to switch the unit off - during writing the display must show the main mask) Bit1 Bit2 Chiller mode switch-on enabling (Send Modbus command with only Bit2 and Bit10 on 1 to switch the unit on in chiller mode - during writing the display must show the main mask) Bit3 Heat pump mode switch-on enabling (Send Modbus command with only Bit3 and Bit11 on 1 to switch the unit on in heat pump mode - during writing the display must show the main mask) Bit4 Bit5 Bit6 Bit7 Bit8 Unit on/off0: stand-by or remote-off 1: chiller or heat pump on (read only) Bit9 On/off from digital input (read only) Bit10 Chiller mode switch-on (Send Modbus command with only Bit2 and Bit10 on 1 to switch the unit on in chiller mode - during writing the display must show the main mask) Bit11 Heat pump mode switch-on (Send Modbus command with only Bit3 and Bit11 on 1 to switch the unit on in heat pump mode - during writing the display must show the main mask) Bit12 Bit13 Bit14 Bit15
1282	Unit status (2nd)	1	
1283	Unit status (3rd)	1	

DIGITAL OUTPUT STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
2049	Relay output status (1st)	1	Bit0 Alarms Bit1 Evaporator water pump / Delivery fan Bit2 Evaporator water pump / Backup delivery fan Bit3 Circuit 1 Anti-freeze, 1st step backup Bit4 Circuit 2 Anti-freeze, 2nd step backup Bit5 Recovery condenser water pump Bit6 Auxiliary recovery condenser water pump relay Bit7 Circuit 1 cycle inversion valve Bit8 Alarm Bit9 Evaporator water pump / Delivery fan Bit10 Evaporator water pump / Backup delivery fan Bit11 Circuit 1 Anti-freeze, 1st step backup Bit12 Circuit 2 Anti-freeze, 2nd step backup Bit13 Recovery condenser water pump Bit14 Auxiliary recovery condenser water pump relay Bit15 Circuit 1 cycle inversion valve

2050	Relay output status (2nd)	1	Bit0 Circuit 2 cycle inversion valve Bit1 Circuit 1 Condensing fans 1 on-off Bit2 Circuit 1 Condensing fans 2 on-off Bit3 Circuit 1 Condensing fans 3 on-off Bit4 Circuit 1 Condensing fans 4 on-off Bit5 Circuit 2 Condensing fans 1 on-off Bit6 Circuit 2 Condensing fans 2 on-off Bit7 Circuit 2 Condensing fans 3 on-off Bit8 Circuit 2 cycle inversion valve Bit9 Circuit 1 Condensing fans 1 on-off Bit10 Circuit 1 Condensing fans 2 on-off Bit11 Circuit 1 Condensing fans 3 on-off Bit12 Circuit 1 Condensing fans 4 on-off Bit13 Circuit 2 Condensing fans 1 on-off Bit14 Circuit 2 Condensing fans 2 on-off Bit15 Circuit 2 Condensing fans 3 on-off
2051	Relay output status (3rd)	1	Bit0 Circuit 2 Condensing fans 4 on-off Bit1 Circuit 1 pump-down Bit2 Circuit 2 pump-down Bit3 Circuit 1 recovery valve Bit4 Circuit 2 recovery valve Bit5 Free-cooling on/off Bit6 Circuit 1 auxiliary Bit7 Circuit 2 auxiliary Bit8 Circuit 2 Condensing fans 4 on-off Bit9 Circuit 1 pump-down Bit10 Circuit 2 pump-down Bit11 Circuit 1 recovery valve Bit12 Circuit 2 recovery valve Bit13 Free-cooling on/off Bit14 Circuit 1 auxiliary Bit15 Circuit 2 auxiliary
2052	Relay output status (4th)	1	Bit0 Compressor 1 intermittent valve Bit1 Compressor 2 intermittent valve Bit2 Compressor 1 liquid injection valve Bit3 Compressor 2 liquid injection valve Bit4 Tank heaters Bit5 Compressor 1 Bit6 Compressor 1 part-winding/star-delta start Bit7 Compressor 1 star centre Bit8 Compressor 1 intermittent valve Bit9 Compressor 2 intermittent valve Bit10 Compressor 1 liquid injection valve Bit11 Compressor 2 liquid injection valve Bit12 Tank heaters Bit13 Compressor 1 Bit14 Compressor 1 part-winding/star-delta start Bit15 Compressor 1 star centre

2053	Relay output status (5th)	1	Bit0 Compressor 1 capacity control 1 Bit1 Compressor 1 capacity control 2 Bit2 Compressor 1 capacity control 3 Bit3 Compressor 1 no-load start with bypass Bit4 Compressor 2 Bit5 Compressor 2 part-winding/star-delta start Bit6 Compressor 2 star centre Bit7 Compressor 2 capacity control 1 Bit8 Compressor 1 capacity control 1 Bit9 Compressor 1 capacity control 2 Bit10 Compressor 1 capacity control 3 Bit11 Compressor 1 no-load start with bypass Bit12 Compressor 2 Bit13 Compressor 2 part-winding/star-delta start Bit14 Compressor 2 star centre Bit15 Compressor 2 capacity control 1
2054	Relay output status (6th)	1	Bit0 Compressor 2 capacity control 2 Bit1 Compressor 2 capacity control 3 Bit2 Compressor 2 no-load start with bypass Bit3 Compressor 3 Bit4 Compressor 3 part-winding/star-delta start Bit5 Compressor 3 star centre Bit6 Compressor 3 capacity reduction 1 Bit7 Compressor 3 capacity reduction 2 Bit8 Compressor 2 capacity reduction 2 Bit9 Compressor 2 capacity reduction 3 Bit10 Compressor 2 no-load start with bypass Bit11 Compressor 3 Bit12 Compressor 3 part-winding/star-delta start Bit13 Compressor 3 star centre Bit14 Compressor 3 capacity control 1 Bit15 Compressor 3 capacity control 2
2055	Relay output status (7th)	1	Bit0 Compressor 3 capacity control 3 Bit1 Compressor 3 no-load start with bypass Bit2 Compressor 4 Bit3 Compressor 4 part-winding/star-delta start Bit4 Compressor 4 star centre Bit5 Compressor 4 capacity control 1 Bit6 Compressor 4 capacity control 2 Bit7 Compressor 4 capacity control 3 Bit8 Compressor 3 capacity control 3 Bit9 Compressor 3 no-load start with bypass Bit10 Compressor 4 Bit11 Compressor 4 part-winding/star-delta start Bit12 Compressor 4 star centre Bit13 Compressor 4 capacity control 1 Bit14 Compressor 4 capacity control 2 Bit15 Compressor 4 capacity control 3
2056	Relay output status (8th)	1	Bit9 Compressor 5 Bit10 Compressor 6

ANALOGUE OUTPUT STATUS (reading)

MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
2305	Circuit 1 Condensing fan output	1	
2306		1	
2307	Circuit 2 Condensing fan output	1	
2308		1	
2309	1st proportional output 0-10 V	1	
2310		1	
2311	2nd proportional output 0-10 V	1	
2312		1	
2313	3rd proportional output 0-10 V	1	
2314		1	
2315	4th proportional output 0-10 V	1	
2316		1	

ALARM STATUS (reading)

MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
3329	Functional alarms (1st)	1	Bit0 Probe 8 alarm Bit1 Probe 9 alarm Bit2 Probe 10 alarm Bit3 Probe 11 alarm Bit4 Probe 12 alarm Bit5 Compressor 1 maintenance Bit6 Compressor 2 maintenance Bit7 Compressor 3 maintenance Bit8 Bit9 Probe 1 alarm Bit10 Probe 2 alarm Bit11 Probe 3 alarm Bit12 Probe 4 alarm Bit13 Probe 5 alarm Bit14 Probe 6 alarm Bit15 Probe 7 alarm
3330	Functional alarms (2nd)	1	Bit0 Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Bit7 Bit8 Compressor 4 maintenance Bit9 Compressor 5 maintenance Bit10 Compressor 6 maintenance Bit11 Pump / delivery fan maintenance Bit12 Evaporator 2 pump maintenance Bit13 Condenser 1 pump maintenance Bit14 Condenser 2 pump maintenance Bit15

3331	Functional alarms (3rd)	1	Bit0 Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Bit7 Bit8 Circuit 1 defrost alarm Bit9 Circuit 2 defrost alarm Bit10 Network frequency Bit11 Low inlet air temperature Bit12 Communication with 1st remote terminal Bit13 Communication with 2nd remote terminal Bit14 Evaporator low outlet temperature Bit15 Evaporator high outlet temperature
3332	Functional alarms (4th)	1	
3333	Functional alarms (5th)	1	
3334	Functional alarms (6th)	1	Bit0 Compressor 1 thermal overload Bit1 Compressor 2 thermal overload Bit2 Compressor 3 thermal overload Bit3 Compressor 4 thermal overload Bit4 Compressor 5 thermal overload Bit5 Compressor 6 thermal overload Bit6 Circuit 1 Condensing fan thermal overload Bit7 Circuit 2 Condensing fan thermal overload Bit8 Bit9 Bit10 Compressor 1 high pressure Bit11 Compressor 2 high pressure Bit12 Compressor 3 high pressure Bit13 Compressor 4 high pressure Bit14 Compressor 5 high pressure Bit15 Compressor 6 high pressure
3335	Functional alarms (7th)	1	Bit0 Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Bit7 Bit8 Evaporator 1 pump thermal overload Bit9 Evaporator 2 pump thermal overload Bit10 Condenser 1 pump thermal overload Bit11 Condenser 2 pump thermal overload Bit12 Circuit 1 high pressure switch tripped Bit13 Circuit 2 high pressure switch tripped Bit14 Circuit 1 high pressure (probe) Bit15 Circuit 2 high pressure (probe)
3336	Functional alarms (8th)	1	
3337	Functional alarms (9th)	1	

3338	Functional alarms (10th)	1	Bit0 Hot side flow switch Bit1 Circuit 1 low pressure switch Bit2 Circuit 2 low pressure switch Bit3 Circuit 1 low pressure (probe) Bit4 Circuit 2 low pressure (probe) Bit5 Circuit 1 pump-down at startup alarm Bit6 Circuit 2 pump-down at startup alarm Bit7 Circuit 1 pump-down at shutdown alarm Bit8 Compressor 1 oil Bit9 Compressor 2 oil Bit10 Compressor 3 oil Bit11 Compressor 4 oil Bit12 Compressor 5 oil Bit13 Compressor 6 oil Bit14 Delivery fan thermal overload Bit15 Cold side flow switch
3339	Functional alarms (11th)	1	Bit0 Compressor 4 discharge high temperature Bit1 Compressor 5 discharge high temperature Bit2 Compressor 6 discharge high temperature Bit3 General alarm from digital input: Bit4 Bit5 Bit6 Bit7 Bit8 Circuit 2 pump-down at shutdown alarm Bit9 Circuit 1 chiller Anti-freeze alarm Bit10 Circuit 2 chiller Anti-freeze alarm Bit11 Circuit 1 pump Anti-freeze alarm Bit12 Circuit 2 pump Anti-freeze alarm Bit13 Compressor 1 discharge high temperature Bit14 Compressor 2 discharge high temperature Bit15 Compressor 3 discharge high temperature

LOAD HOURS (read-write)

MODBUS ADDRESS(DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
3585	Compressor 1 operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	
3586	Compressor 2 operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	
3587	Compressor 3 operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	
3588	Compressor 4 operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	
3589	Compressor 5 operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	
3590	Compressor 6 operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	
3591	Evaporator 1 pump operating hours Set to 0 to reset -during writing the main mask must be displayed)	1	
3592	Evaporator 2 pump operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	

3593	Condenser 1 pump operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	
3594	Condenser 2 pump operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	

ALARMS TRIPPED PER HOUR (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
3841	Compressor 1 oil	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3842	Compressor 2 oil	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3843	Compressor 3 oil	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3844	Compressor 4 oil	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3845	Compressor 5 oil	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3846	Compressor 6 oil	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3847	Delivery fan thermal overload	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3848	Cold side flow switch	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3849	Hot side flow switch	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3850	Circuit 1 low temperature/pressure from digital input	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3851	Circuit 2 low temperature/pressure from digital input	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3852	Circuit 1 low temperature/pressure from probe	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3853	Circuit 2 low temperature/pressure from probe	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3854	Circuit 1 pump-down at startup	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3855	Circuit 2 pump-down at startup	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3856	Circuit 1 pump-down at shutdown	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3857	Circuit 2 pump-down at shutdown	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3858	Circuit 1 chiller Anti-freeze	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3859	Circuit 2 chiller Anti-freeze	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3860	Circuit 1 pump Anti-freeze	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3861	Circuit 2 pump Anti-freeze	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3862	Compressor 1 discharge temperature	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3863	Compressor 2 discharge temperature	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3864	Compressor 3 discharge temperature	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3865	Compressor 4 discharge temperature	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3866	Compressor 5 discharge temperature	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3867	Compressor 6 discharge temperature	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold
3868	General alarm	1	Bit15-Bit8: number of trips recorded; Bit0-Bit7: alarm threshold

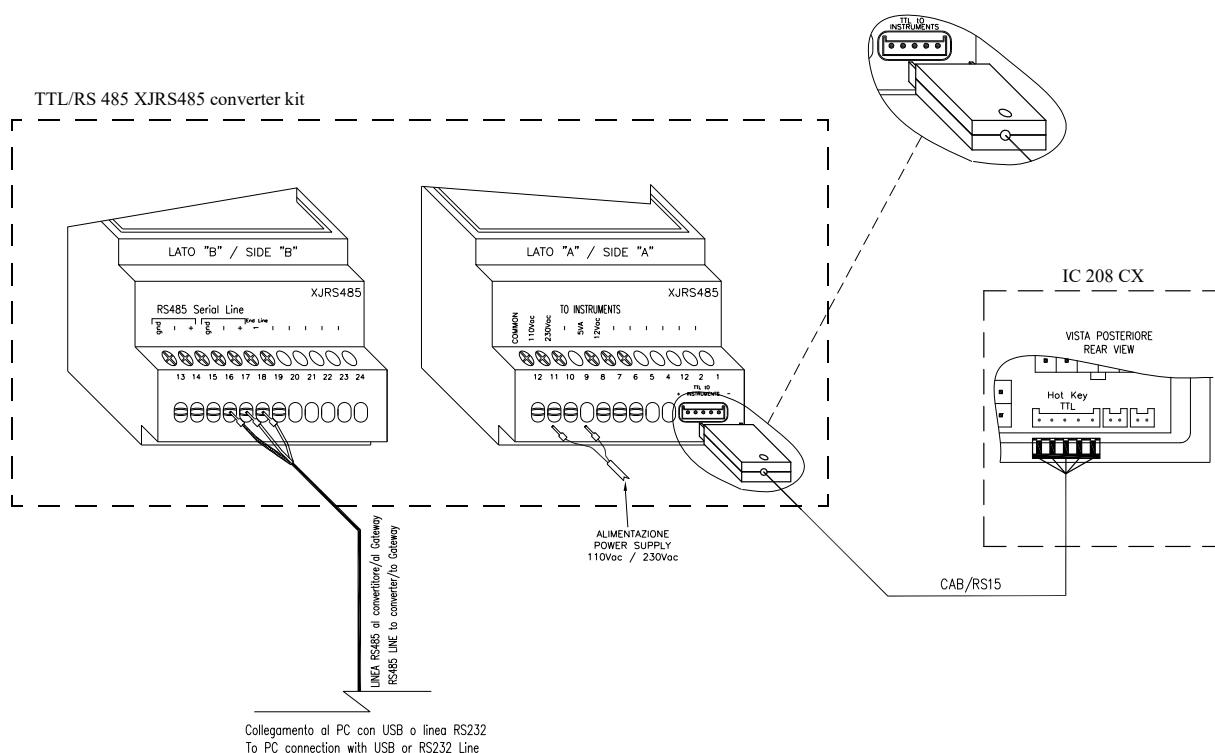
IC208CX DRY



5. MODBUS-RTU

5.1 Connection drawing

The drawing below shows how to connect the unit to a Modbus network:



5.2 Configuration

Communication is configured as follows:

Baud Rate = 9600 bps

Data Length = 8 bit

Parity = None

Stop Bit = 1

5.3 How to set the device address

Press **SET** +  buttons for 3 sec. to enter the programming procedure, then press **SET** until reaching **CF64** parameter.

Use  and  buttons to modify the value.

Press **SET** again to store the new value.

5.4 Variables

DEVICE DATA				
REGISTER	MODBUS ADDRESS (DEC)	DESCRIPTION	SIZE (word)	NOTE
0	1	Product family code Firmware release	1	MSByte = Product family code LSByte = Firmware release
5	6	FW version date	1	Day = bit15 / bit11 Month = bit10 / bit7 Year = bit6 / bit0

VALUE AND PROBE STATUS IC200CX				
REGISTER	MODBUS ADDRESS (DEC)	DESCRIPTION	SIZE (word)	NOTE
256	257	Probe 1 value	1	
258	259	Probe 2 value	1	
260	261	Probe 3 value	1	
262	263	Probe 4 value	1	
264	265	Probe 5 value	1	
266	267	Probe 6 value	1	
268	269	Remote keyboard 1 probe value	1	
270	271	Remote keyboard 2 probe value	1	
272	273	I/O expansion probe 1 value	1	
274	275	I/O expansion probe 2 value	1	
276	277	I/O expansion probe 3 value	1	
278	279	I/O expansion probe 4 value	1	
280	281	I/O expansion probe 5 value	1	
282	283	I/O expansion probe 6 value	1	
284	285	I/O expansion probe 7 value	1	
286	287	I/O expansion probe 8 value	1	
288	289	Expansion valve probe 1 value	1	
290	291	Expansion valve probe 2 value	1	
292	293	Expansion valve probe 3 value		
294	295	Expansion valve probe 4 value		

DIGITAL INPUT STATUS			
MSB	RESOURCE STATUS	LSB	RESOURCE STATUS
Register 512 / ModBus address (DEC) 513			
bit0	Remote Heat Pump	bit0	High pressure switch circuit 2
bit1	Remote OFF	bit1	Low pressure switch circuit 1
bit2	Remote chiller	bit2	Low pressure switch circuit 2
bit3	Flow switch/ Supply fan overload	bit3	Compressor 1 high pressure
bit4	Flow switch of heated side	bit4	Compressor 2 high pressure
bit5	Antifreeze alarm circuit 1	bit5	Compressor 3 high pressure
bit6	Antifreeze alarm circuit 2	bit6	Compressor 4 high pressure
bit7	High pressure switch circuit 1	bit7	Not used
Register 513 / ModBus address (DEC) 514			
bit0	Not used	bit0	Condenser fan overload of circuit 2
bit1	Compressor 1 overload	bit1	Condenser fan overload of circuit 1 and 2 (common)
bit2	Compressor 2 overload	bit2	Water pump overload of evaporator 1
bit3	Compressor 3 overload	bit3	Water support pump overload of evaporator
bit4	Compressor 4 overload	bit4	Water pump overload of condenser 1
bit5	Not used	bit5	Water support pump overload of condenser
bit6	Not used	bit6	Recovery request for circuit 1
bit7	Condenser fan overload of circuit 1	bit7	Recovery request for circuit 2

Register 514 / ModBus address (DEC) 515			
bit0	End defrost circuit 1	bit0	Not used
bit1	End defrost circuit 2	bit1	Pump down pressure switch of circuit 1
bit2	Energy Saving	bit2	Pump down pressure switch of circuit 2
bit3	Pressure switch / oil level compressor 1	bit3	Generic alarm from digital input
bit4	Pressure switch / oil level compressor 2	bit4	Generic alarm 2 from digital input
bit5	Pressure switch / oil level compressor 3	bit5	Automatic working mode (time table)
bit6	Pressure switch / oil level compressor 4	bit6	Only ventilation (supply fan)
bit7	Not used	bit7	Regulation request (condensing unit)
Register 515 / ModBus address (DEC) 516			
bit0	Cooling request (condensing unit)	bit0	Step 8 request (condensing unit)
bit1	Heating request (condensing unit)	bit1	Step 9 request (condensing unit)
bit2	Step 2 request (condensing unit)	bit2	Step 10 request (condensing unit)
bit3	Step 3 request (condensing unit)	bit3	Step 11 request (condensing unit)
bit4	Step 4 request (condensing unit)	bit4	Step 12 request (condensing unit)
bit5	Step 5 request (condensing unit)	bit5	Step 13 request (condensing unit)
bit6	Step 6 request (condensing unit)	bit6	Step 14 request (condensing unit)
bit7	Step 7 request (condensing unit)	bit7	Step 15 request (condensing unit)
Register 516 / ModBus address (DEC) 517			
bit0	Step 16 request (condensing unit)	bit0	Sanitary water priority
bit1	Sanitary flow switch	bit1	Free cooling water pump switch
bit2	Solar panel flow switch	bit2	Expansion valve alarm circuit 1
bit3	Only sanitary hot water	bit3	Expansion valve alarm circuit 2
bit4	Overload sanitary water heaters	bit4	Condensate drain alarm
bit5	Overload sanitary water pump	bit5	Antifreeze alarm for all circuits
bit6	Enable sanitary water second set point	bit6	
bit7	Supply voltage sequence phase failure	bit7	

READING/WRTING MACHINE STATUS			
Register 1280 / ModBus address (DEC) 1281			
MSByte	STATUS MODIFICATION	LSByte	MODIFICATION ENABLED
bit0	Device status Reading: bit=0 if device in stand-by or remote off, bit=1 if device on in chiller or heat pump Writing: allowed only device switching off setting bit=0	bit0	Enable device switch off Writing: only switch off is allowed; set bit=0 to switch off the device
bit1	Remote ON/OFF status Only reading: bit =1 device on, bit =0 device off by digital input		
bit2	Chiller status Reading: bit =1 if the device is on in chiller Writing: set bit =1 to switch on the device in chiller	bit2	Chiller status Reading: if bit =1 device on in chiller Writing: set bit=1 to switch on the device in chiller
bit3		bit3	

READING/WRTING SETPOINT			
REGISTER	DESCRIPTION	SIZE (word)	NOTE
Register 768 / ModBus address (DEC) 769	Chiller Set point	1	Reading
Register 779 / ModBus address (DEC) 780	Dew point set	1	Reading / Writing

RELAY STATUS			
MSByte	RELAY STATUS	LSByte	RELAY STATUS
Register 2048 / ModBus address (DEC) 2049			
bit0	Alarm	bit0	Alarm
bit1	Water pump of the evaporator / supply fan	bit1	Water pump of the evaporator / supply fan
bit2	Supply water pump of the evaporator / support supply fan	bit2	Supply water pump of the evaporator / support supply fan
bit3	Antifreeze heater circuit 1 (first step)	bit3	Antifreeze heater circuit 1 (first step)
bit4	Antifreeze heater circuit 2 (second step)	bit4	Antifreeze heater circuit 2 (second step)
bit5	Condenser water pump	bit5	Condenser water pump
bit6	Condenser support water pump	bit6	Condenser support water pump
bit7	Reversing valve circuit 1	bit7	Reversing valve circuit 1
Register 2049/ModBus address (DEC) 2050			
bit0	Reversing valve circuit 2	bit0	Reversing valve circuit 2
bit1	Condensing fan 1 of the circuit 1	bit1	Condensing fan 1 of the circuit 1
bit2	Condensing fan 2 of the circuit 1	bit2	Condensing fan 2 of the circuit 1
bit3	Condensing fan 3 of the circuit 1	bit3	Condensing fan 3 of the circuit 1
bit4	Condensing fan 4 of the circuit 1	bit4	Condensing fan 4 of the circuit 1
bit5	Condensing fan 1 of the circuit 2	bit5	Condensing fan 1 of the circuit 2
bit6	Condensing fan 2 of the circuit 2	bit6	Condensing fan 2 of the circuit 2
bit7	Condensing fan 3 of the circuit 2	bit7	Condensing fan 3 of the circuit 2
Register 2050 / ModBus address (DEC) 2051			
bit0	Condensing fan 4 of the circuit 2	bit0	Condensing fan 4 of the circuit 2
bit1	Pump_down valve circuit 1	bit1	Pump_down valve circuit 1
bit2	Pump_down valve circuit 2	bit2	Pump_down valve circuit 2
bit3	Recovery valve circuit 1	bit3	Recovery valve circuit 1
bit4	Recovery valve circuit 2	bit4	Recovery valve circuit 2
bit5	Free cooling valve	bit5	Free cooling valve
bit6	Auxiliary output 1	bit6	Auxiliary output 1
bit7	Auxiliary output 2	bit7	Auxiliary output 2
Register 2051/ModBus address (DEC) 2052			
bit0	Intermittent valve compressor 1	bit0	Intermittent valve compressor 1
bit1	Intermittent valve compressor 2	bit1	Intermittent valve compressor 2
bit2	Valve for liquid injection compressor 1	bit2	Valve for liquid injection compressor 1
bit3	Valve for liquid injection compressor 2	bit3	Valve for liquid injection compressor 2
bit4	Valve 1 for sanitary water	bit4	Valve 1 for sanitary water
bit5	Valve 2 for sanitary water	bit5	Valve 2 for sanitary water
bit6	Heater 1 for sanitary water	bit6	Heater 1 for sanitary water
bit7	Heater 2 for sanitary water	bit7	Heater 2 for sanitary water
Register 2052/ModBus address (DEC) 2053			
bit0	Heater 3 for sanitary water	bit0	Heater 3 for sanitary water
bit1	Solar panel water pump	bit1	Solar panel water pump
bit2	Solar panel valve	bit2	Solar panel valve
bit3	Sanitary water pump	bit3	Sanitary water pump
bit4	Hybrid exchanger 1 circuit 1	bit4	Hybrid exchanger 1 circuit 1
bit5	Hybrid exchanger 2 circuit 1	bit5	Hybrid exchanger 2 circuit 1
bit6	Hybrid exchanger 1 circuit 2	bit6	Hybrid exchanger 1 circuit 2
bit7	Hybrid exchanger 2 circuit 2	bit7	Hybrid exchanger 2 circuit 2
Register 2053/ModBus address (DEC) 2054			
bit0	Cooling/heating circuit 1	bit0	Cooling/heating circuit 1
bit1	Cooling/heating circuit 2	bit1	Cooling/heating circuit 2
bit2	Defrost circuit 1	bit2	Defrost circuit 1
bit3	Defrost circuit 2	bit3	Defrost circuit 2
bit4	Circuit 1 on power request (cooling or heating)	bit4	Circuit 1 on power request (cooling or heating)
bit5	Circuit 2 on power request (cooling or heating)	bit5	Circuit 2 on power request (cooling or heating)
bit6	Sanitary water circuit on power request	bit6	Sanitary water circuit on power request
bit7	Stand-by or remote off	bit7	Stand-by or remote off

RELAY STATUS			
MSByte	RELAY STATUS	LSByte	RELAY STATUS
Register 2054/ModBus address (DEC) 2055			
bit0	Solenoid valve circuit 1	bit0	Solenoid valve circuit 1
bit1	Solenoid valve circuit 2	bit1	Solenoid valve circuit 2
bit2	Ready to start	bit2	Ready to start
bit3	Condensate drain 1	bit3	Condensate drain 1
bit4	Condensate drain 2	bit4	Condensate drain 2
bit5	Compressor 1	bit5	Compressor 1
bit6	Part-winding coil compressor 1	bit6	Part-winding coil compressor 1
bit7	Capacity step valve 1 compressor 1	bit7	Capacity step valve 1 compressor 1
Register 2055/ModBus address (DEC) 2056			
bit0	Capacity step valve 2 compressor 1	bit0	Capacity step valve 2 compressor 1
bit1	Capacity step valve 3 compressor 1	bit1	Capacity step valve 3 compressor 1
bit2	By-pass valve compressor 1	bit2	By-pass valve compressor 1
bit3	Compressor 2	bit3	Compressor 2
bit4	Part-winding coil compressor 2	bit4	Part-winding coil compressor 2
bit5	Capacity step valve 1 compressor 2	bit5	Capacity step valve 1 compressor 2
bit6	Capacity step valve 2 compressor 2	bit6	Capacity step valve 2 compressor 2
bit7	Capacity step valve 3 compressor 2	bit7	Capacity step valve 3 compressor 2
Register 2056/ModBus address (DEC) 2057			
bit0	By-pass valve compressor 2	bit0	By-pass valve compressor 2
bit1	Compressor 3	bit1	Compressor 3
bit2	Part-winding coil compressor 3	bit2	Part-winding coil compressor 3
bit3	Capacity step valve 1 compressor 3	bit3	Capacity step valve 1 compressor 3
bit4	Capacity step valve 2 compressor 3	bit4	Capacity step valve 2 compressor 3
bit5	Capacity step valve 3 compressor 3	bit5	Capacity step valve 3 compressor 3
bit6	By-pass valve compressor 3	bit6	By-pass valve compressor 3
bit7	Compressor 4	bit7	Compressor 4
Register 2057/ModBus address (DEC) 2058			
bit0	Part-winding coil compressor 4	bit0	Part-winding coil compressor 4
bit1	Capacity step valve 1 compressor 4	bit1	Capacity step valve 1 compressor 4
bit2	Capacity step valve 2 compressor 4	bit2	Capacity step valve 2 compressor 4
bit3	Capacity step valve 3 compressor 4	bit3	Capacity step valve 3 compressor 4
bit4	By-pass valve compressor 4	bit4	By-pass valve compressor 4
bit5		bit5	
bit6		bit6	
bit7		bit7	

ANALOGUE OUTPUTS STATUS (reading)				
Register	ModBus address (DEC)	DESCRIPTION	SIZE (word)	NOTE
2304	2305	Analogue output 1	1	0 -100 %
2305	2306	Analogue output 2	1	0 -100 %
2306	2307	Analogue output 3	1	0 -100 %
2307	2308	Analogue output 4	1	0 -100 %
2308	2309	I/O Expansion Analogue output 1	1	0 -100 %
2309	2310	I/O Expansion Analogue output 2	1	0 -100 %
2310	2311	I/O Expansion Analogue output 3	1	0 -100 %

ALARM STATUS (reading)			
MSB	DESCRIPTION	LSB	DESCRIPTION
Register 3328/ModBus address (DEC) 3329			
bit0	not valid timetable	bit0	Not used
bit1	probe 1 alarm	bit1	Not used
bit2	probe 2 alarm	bit2	Not used
bit3	probe 3 alarm	bit3	Remote keyboard 1 probe alarm
bit4	probe 4 alarm	bit4	Remote keyboard 2 probe alarm
bit5	probe 5 alarm	bit5	Compressor 1 maintenance warning
bit6	probe 6 alarm	bit6	Compressor 2 maintenance warning
bit7	Not used	bit7	Compressor 3 maintenance warning
Register 3329/ModBus address (DEC) 3330			
bit0	Compressor 4 maintenance warning	bit0	Solar panel water pump maintenance warning
bit1	Not used	bit1	Free cooling water pump maintenance warning
bit2	Not used	bit2	Configuration alarm 2
bit3	Water pump / supply fan maintenance warning	bit3	Configuration alarm 3
bit4	Evaporator 2 pump maintenance warning	bit4	Configuration alarm 4
bit5	Condenser 1 pump maintenance warning	bit5	Configuration alarm 5
bit6	Condenser 2 pump maintenance warning	bit6	Configuration alarm 6
bit7	Sanitary water pump maintenance warning	bit7	Configuration alarm 7
Register 3330/ModBus address (DEC) 3331			
bit0	Configuration alarm 8	bit0	1st remote keyboard communication failure
bit1	Configuration alarm 9	bit1	2nd remote keyboard communication failure
bit2	Configuration alarm 10	bit2	Free cooling configuration alarm
bit3	Configuration alarm 11	bit3	Sanitary water configuration alarm
bit4	Defrost alarm circuit 1	bit4	Power supply phase sequence alarm
bit5	Defrost alarm circuit 2	bit5	Configuration 1 alarm
bit6	Not used	bit6	Configuration 14 alarm
bit7	Low inlet air temperature	bit7	I/O expansion board communication alarm
Register 3331/ModBus address (DEC) 3332			
bit0	Expansion valve communication alarm	bit0	I/O expansion board probe 8 error
bit1	I/O expansion board probe 1 error	bit1	Expansion valve probe 1 error
bit2	I/O expansion board probe 2 error	bit2	Expansion valve probe 2 error
bit3	I/O expansion board probe 3 error	bit3	Expansion valve probe 3 error
bit4	I/O expansion board probe 4 error	bit4	Expansion valve probe 4 error
bit5	I/O expansion board probe 5 error	bit5	Expansion valve alarm circuit 1
bit6	I/O expansion board probe 6 error	bit6	Expansion valve alarm circuit 2
bit7	I/O expansion board probe 7 error	bit7	Not used
Register 3332/ModBus address (DEC) 3333			
bit0	Condensate drain	bit0	Condensate drain
bit1	Air inlet high temperature	bit1	Air inlet high temperature
bit2	Air outlet high temperature	bit2	Air outlet high temperature
bit3	Dewpont high temperature	bit3	Dewpont high temperature
bit4	Not used	bit4	Not used
bit5	Not used	bit5	Not used
bit6	Not used	bit6	Not used
bit7	Not used	bit7	Not used

ALARM STATUS (reading)			
MSB	DESCRIPTION	LSB	DESCRIPTION
Register 3333/ModBus address (DEC) 3334			
bit0	EEPROM alarm	bit0	Compressor 1 overload
bit1	Faulty clock	bit1	Compressor 2 overload
bit2	Compressor 1 high pressure	bit2	Compressor 3 overload
bit3	Compressor 2 high pressure	bit3	Compressor 4 overload
bit4	Compressor 3 high pressure	bit4	Not used
bit5	Compressor 4 high pressure	bit5	Not used
bit6	Not used	bit6	Condensing fan overload circuit 1
bit7	Not used	bit7	Condensing fan overload circuit 2
Register 3334/ModBus address (DEC) 3335			
bit0	Evaporator 1 pump overload	bit0	
bit1	Evaporator 2 pump overload	bit1	
bit2	Condenser 1 pump overload	bit2	
bit3	Condenser 2 pump overload	bit3	
bit4	Supply fan overload	bit4	
bit5	Contemporary heating and coling request (condensing unit)	bit5	
bit6		bit6	
bit7		bit7	
Register 3337/ModBus address (DEC) 3338			
bit0	Compressor 1 oil level	bit0	Condenser flow switch
bit1	Compressor 2 oil level	bit1	Low pressure switch circuit 1 (digital input)
bit2	Compressor 3 oil level	bit2	Low pressure switch circuit 2 (digital input)
bit3	Compressor 4 oil level	bit3	Low pressure switch circuit 1 (probe)
bit4	Not used	bit4	Low pressure switch circuit 2 (probe)
bit5	Not used	bit5	Pump-down circuit 1 alarm
bit6	High inlet temperature	bit6	Pump-down circuit 2 alarm
bit7	Evaporator flow switch	bit7	Pump-down at stop circuit 1 alarm
Register 3338/ModBus address (DEC) 3339			
bit0	Pump-down at stop circuit 2 alarm	bit0	High temperature compressor 4 discharge
bit1	Chiller antifreeze heater circuit 1	bit1	Not used
bit2	Chiller antifreeze heater circuit 2	bit2	Not used
bit3	Heat Pump antifreeze heater circuit 1 alarm	bit3	General alarm 1
bit4	Heat Pump antifreeze heater circuit 2 alarm	bit4	General alarm 2
bit5	High temperature compressor 1 discharge	bit5	High pressure switch circuit 1 (digital input)
bit6	High temperature compressor 2 discharge	bit6	High pressure switch circuit 2 (digital input)
bit7	High temperature compressor 3 discharge	bit7	High pressure switch circuit 1 (probe)
Register 3339/ModBus address (DEC) 3340			
bit0	High pressure switch circuit 2 (probe)	bit0	Air inlet low temperature pre-alarm
bit1	Sanitary water flow switch	bit1	Air outlet low temperature pre-alarm
bit2	Solar panel flow switch	bit2	Dewpoint low temperature pre-alarm
bit3	Overload sanitary water heaters	bit3	Air inlet low temperature alarm
bit4	Overload sanitary water pump	bit4	Air outlet low temperature alarm
bit5	Compressor 1 oil differential pressure switch	bit5	Dewpoint low temperature alarm
bit6	Compressor 2 oil differential pressure switch	bit6	
bit7	Free cooling water pump switch	bit7	

WORKING HOURS (reading - writing)

Register	ModBus address (DEC)	DESCRIPTION	SIZE (word)	NOTE
3584	3585	Compressor 1 running hours	1	
3585	3586	Compressor 2 running hours	1	
3586	3587	Compressor 3 running hours	1	
3587	3588	Compressor 4 running hours	1	
3588	3589		1	
3589	3590		1	
3590	3591	Evaporator water pump 1 or Supply fan running hours	1	
3591	3592	Evaporator water pump 2 or Supply fan running hours	1	
3592	3593	Condenser pump 1 running hours	1	
3593	3594	Condenser pump 2 running hours	1	
3594	3595	Sanitary water pump running hours	1	
3595	3596	Solar panel running hours	1	
3596	3597	Free cooling water pump running hours	1	

ENERGY SAVING

Register	ModBus address (DEC)	DESCRIPTION	SIZE (word)	NOTE
4096	4097	Energy saving percentage	1	

NUMBER OF ACTIVE ALARMS PER HOUR**NUMBER OF ACTIVE ALARMS PER HOUR (reading)**

Register 3840	ModBus address (DEC) 3841	COMPRESSOR 1 OIL LEVEL		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3841	ModBus address (DEC) 3842	COMPRESSOR 2 OIL LEVEL		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3842	ModBus address (DEC) 3843	COMPRESSOR 3 OIL LEVEL		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3843	ModBus address (DEC) 3844	COMPRESSOR 4 OIL LEVEL		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3844	ModBus address (DEC) 3845			
MSByte		N° of activation	LSByte	Alarm threshold
Register 3845	ModBus address (DEC) 3846			
MSByte		N° of activation	LSByte	Alarm threshold
Register 3846	ModBus address (DEC) 3847	HIGH TEMPERATURE (CONFIGURABLE PROBE)		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3847	ModBus address (DEC) 3848	NOT USED		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3848	ModBus address (DEC) 3849	NOT USED		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3849	ModBus address (DEC) 3850	LOW TEMPERATURE /PRESSURE CIRCUIT 1 (Digital input)		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3850	ModBus address (DEC) 3851	LOW TEMPERATURE /PRESSURE CIRCUIT 2 (Digital input)		
MSByte		N° of activation	LSByte	Alarm threshold

Register 3851	ModBus address (DEC) 3852	LOW TEMPERATURE /PRESSURE CIRCUIT 1 (Probe)		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3852	ModBus address (DEC) 3853	LOW TEMPERATURE /PRESSURE CIRCUIT 2 (Probe)		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3853	ModBus address (DEC) 3854	PUMP_DOWN AT THE START-UP CIRCUIT 1		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3854	ModBus address (DEC) 3855	PUMP_DOWN AT THE START-UP CIRCUIT 2		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3855	ModBus address (DEC) 3856	PUMP_DOWN IN SWITCHING OFF CIRCUIT 1		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3856	ModBus address (DEC) 3857	PUMP_DOWN IN SWITCHING OFF CIRCUIT 2		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3857	ModBus address (DEC) 3858	ANTIFREEZE CHILLER CIRCUIT 1		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3858	ModBus address (DEC) 3859	ANTIFREEZE CHILLER CIRCUIT 2		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3859	ModBus address (DEC) 3860	ANTIFREEZE HEAT PUMP CIRCUIT 1		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3860	ModBus address (DEC) 3861	ANTIFREEZE HEAT PUMP CIRCUIT 2		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3861	ModBus address (DEC) 3862	COMPRESSOR 1 DISCHARGE TEMPERATURE		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3862	ModBus address (DEC) 3863	COMPRESSOR 2 DISCHARGE TEMPERATURE		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3863	ModBus address (DEC) 3864	COMPRESSOR 3 DISCHARGE TEMPERATURE		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3864	ModBus address (DEC) 3865	COMPRESSOR 4 DISCHARGE TEMPERATURE		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3865	ModBus address (DEC) 3866			
MSByte		N° of activation	LSByte	Alarm threshold
Register 3866	ModBus address (DEC) 3867			
MSByte		N° of activation	LSByte	Alarm threshold
Register 3867	ModBus address (DEC) 3868	GENERAL ALARM 1		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3868	ModBus address (DEC) 3869	GENERAL ALARM 2		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3869	ModBus address (DEC) 3870	CIRCUIT 1 HIGH PRESSURE ALARM (DIGITAL INPUT)		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3870	ModBus address (DEC) 3871	CIRCUIT 2 HIGH PRESSURE ALARM (DIGITAL INPUT)		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3871	ModBus address (DEC) 3872	CIRCUIT 1 HIGH PRESSURE ALARM (PROBE)		
MSByte		N° of activation	LSByte	Alarm threshold

Register 3872	ModBus address (DEC) 3873	CIRCUIT 2 HIGH PRESSURE ALARM (PROBE)		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3873	ModBus address (DEC) 3874	NOT USED		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3874	ModBus address (DEC) 3875	NOT USED		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3875	ModBus address (DEC) 3876	OVERLOAD SANITARY WATER HEATERS		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3876	ModBus address (DEC) 3877	OVERLOAD SANITARY WATER PUMP		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3877	ModBus address (DEC) 3878	PRESSURE OIL DIFFERENTIAL SWITCH CIRCUIT 1 OR COMPRESSOR 1		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3878	ModBus address (DEC) 3879	PRESSURE OIL DIFFERENTIAL SWITCH CIRCUIT 1 OR COMPRESSOR 2		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3879	ModBus address (DEC) 3880	NOT USED		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3880	ModBus address (DEC) 3881	OVERLOAD COMPRESSOR 1		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3881	ModBus address (DEC) 3882	OVERLOAD COMPRESSOR 2		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3882	ModBus address (DEC) 3883	OVERLOAD COMPRESSOR 3		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3883	ModBus address (DEC) 3884	OVERLOAD COMPRESSOR 4		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3884	ModBus address (DEC) 3885			
MSByte		N° of activation	LSByte	Alarm threshold
Register 3885	ModBus address (DEC) 3886			
MSByte		N° of activation	LSByte	Alarm threshold

IC208CX FC4ALL - FC4TAE



6. MODBUS-RTU

6.1 Connection drawing

For the connection diagram, refer to the wiring diagram of the FC4ALL and FC4TAE modules.

6.2 Configuration

Communication is configured as follows:

Baud Rate = 9600 bps

Data Length = 8 bit

Parity = None

Stop Bit = 1

6.3 How to set the device address

Press **SET** +  buttons for 3 sec. to enter the programming procedure, then press **SET** until reaching CF42 parameter.

Use  and  buttons to modify the value.

Press **SET** again to store the new value.

6.4 Variables

PROBE STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
277	Temperature probe at water inlet	1	
278	Ambient temperature probe	1	
279	Temperature probe at water outlet	1	
280	Status of temperature probe at water inlet	1	
281	Status of ambient temperature probe	1	
282	Status of temperature probe at water outlet	1	

DIGITAL VARIABLES (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE Bit	NOTE
533	Communication blink	1	
534	Fan overload alarm	1	
535	Freecooling probe alarm	1	
536	Ambient probe alarm	1	
537	High temperature alarm for freecooling probe	1	
538	Low temperature alarm for freecooling probe	1	
539	Pump alarm	1	
540	Pressure switch alarm	1	
541	Alarm for water outlet probe	1	
542	High temperature alarm for water outlet probe	1	
543	Low temperature alarm for water outlet probe	1	

544	Available alarm	1	
545	Pump relay status	1	
546	Alarm relay status	1	

DIGITAL VARIABLES (reading-writing)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE Bit	NOTE
518	On/Off	1	

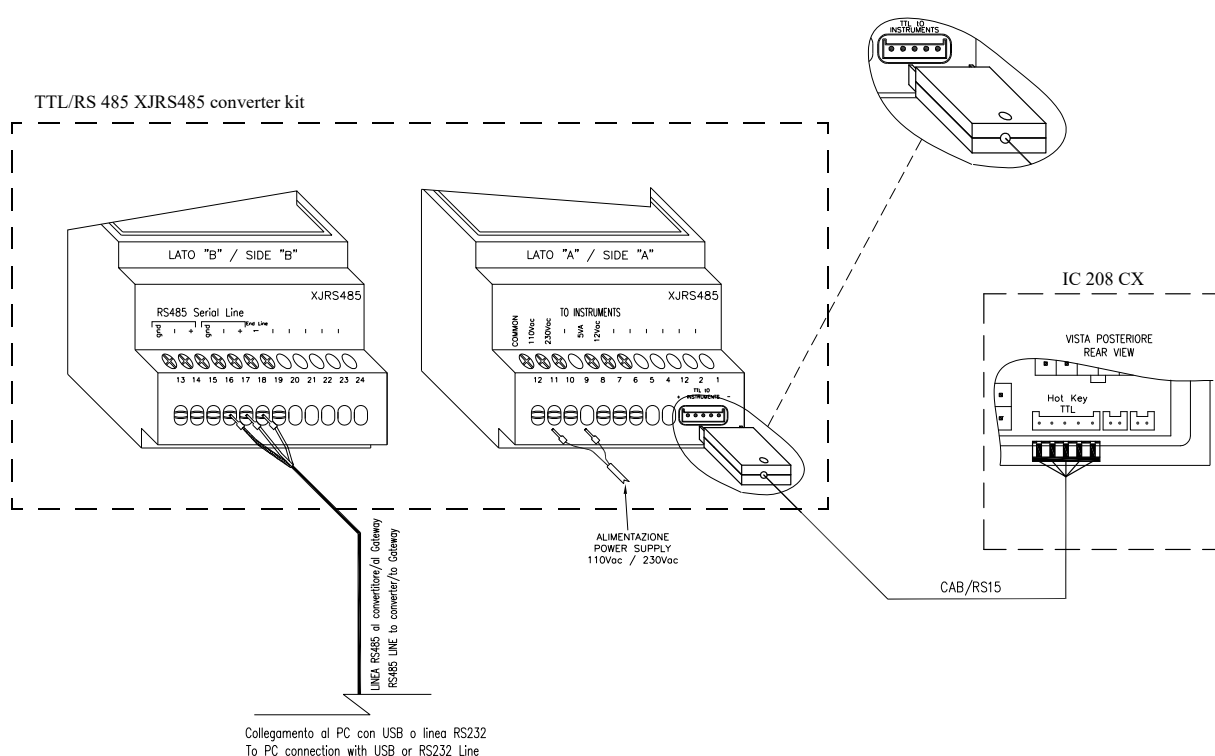
IC208CX TAT - HTAT - MCTAT - TAEvo Tech



7. MODBUS-RTU

7.1 Connection drawing

The drawing below shows how to connect the unit to a Modbus network:



7.2 Configuration

Communication is configured as follows:

Baud Rate = 9600 bps

Data Length = 8 bit

Parity = None

Stop Bit = 1

7.3 How to set the device address

Press **SET** +  buttons for 3 sec. to enter the programming procedure, then press **SET** until reaching CF64 parameter.

Use  and  buttons to modify the value.

Press **SET** again to store the new value.

7.4 Variables

DEVICE DATA				
REGISTER	MODBUS ADDRESS (DEC)	DESCRIPTION	SIZE (word)	NOTE
0	1	Product family code Firmware release	1	MSByte = Product family code LSByte = Firmware release
5	6	FW version date	1	Day = bit15 / bit11 Month = bit10 / bit7 Year = bit6 / bit0

VALUE AND PROBE STATUS IC200CX				
REGISTER	MODBUS ADDRESS (DEC)	DESCRIPTION	SIZE (word)	NOTE
256	257	Probe 1 value	1	
258	259	Probe 2 value	1	
260	261	Probe 3 value	1	
262	263	Probe 4 value	1	
264	265	Probe 5 value	1	
266	267	Probe 6 value	1	
268	269	Remote keyboard 1 probe value	1	
270	271	Remote keyboard 2 probe value	1	
272	273	I/O expansion probe 1 value	1	
274	275	I/O expansion probe 2 value	1	
276	277	I/O expansion probe 3 value	1	
278	279	I/O expansion probe 4 value	1	
280	281	I/O expansion probe 5 value	1	
282	283	I/O expansion probe 6 value	1	
284	285	I/O expansion probe 7 value	1	
286	287	I/O expansion probe 8 value	1	
288	289	Expansion valve probe 1 value	1	
290	291	Expansion valve probe 2 value	1	
292	293	Expansion valve probe 3 value	1	
294	295	Expansion valve probe 4 value	1	

DIGITAL INPUT STATUS			
MSB	RESOURCE STATUS	LSB	RESOURCE STATUS
Register 512 / ModBus address (DEC) 513			
bit0	Remote Heat Pump	bit0	High pressure switch circuit 2
bit1	Remote OFF	bit1	Low pressure switch circuit 1
bit2	Remote chiller	bit2	Low pressure switch circuit 2
bit3	Flow switch/ Supply fan overload	bit3	Compressor 1 high pressure
bit4	Flow switch of heated side	bit4	Compressor 2 high pressure
bit5	Antifreeze alarm circuit 1	bit5	Compressor 3 high pressure
bit6	Antifreeze alarm circuit 2	bit6	Compressor 4 high pressure
bit7	High pressure switch circuit 1	bit7	Not used
Register 513 / ModBus address (DEC) 514			
bit0	Not used	bit0	Condenser fan overload of circuit 2
bit1	Compressor 1 overload	bit1	Condenser fan overload of circuit 1 and 2 (common)
bit2	Compressor 2 overload	bit2	Water pump overload of evaporator 1
bit3	Compressor 3 overload	bit3	Water support pump overload of evaporator
bit4	Compressor 4 overload	bit4	Water pump overload of condenser 1
bit5	Not used	bit5	Water support pump overload of condenser
bit6	Not used	bit6	Recovery request for circuit 1
bit7	Condenser fan overload of circuit 1	bit7	Recovery request for circuit 2

Register 514 / ModBus address (DEC) 515			
bit0	End defrost circuit 1	bit0	Not used
bit1	End defrost circuit 2	bit1	Pump down pressure switch of circuit 1
bit2	Energy Saving	bit2	Pump down pressure switch of circuit 2
bit3	Pressure switch / oil level compressor 1	bit3	Generic alarm from digital input
bit4	Pressure switch / oil level compressor 2	bit4	Generic alarm 2 from digital input
bit5	Pressure switch / oil level compressor 3	bit5	Automatic working mode (time table)
bit6	Pressure switch / oil level compressor 4	bit6	Only ventilation (supply fan)
bit7	Not used	bit7	Regulation request (condensing unit)
Register 515 / ModBus address (DEC) 516			
bit0	Cooling request (condensing unit)	bit0	Step 8 request (condensing unit)
bit1	Heating request (condensing unit)	bit1	Step 9 request (condensing unit)
bit2	Step 2 request (condensing unit)	bit2	Step 10 request (condensing unit)
bit3	Step 3 request (condensing unit)	bit3	Step 11 request (condensing unit)
bit4	Step 4 request (condensing unit)	bit4	Step 12 request (condensing unit)
bit5	Step 5 request (condensing unit)	bit5	Step 13 request (condensing unit)
bit6	Step 6 request (condensing unit)	bit6	Step 14 request (condensing unit)
bit7	Step 7 request (condensing unit)	bit7	Step 15 request (condensing unit)
Register 516 / ModBus address (DEC) 517			
bit0	Step 16 request (condensing unit)	bit0	Sanitary water priority
bit1	Sanitary flow switch	bit1	Free cooling water pump switch
bit2	Solar panel flow switch	bit2	Expansion valve alarm circuit 1 (*)
bit3	Only sanitary hot water	bit3	Expansion valve alarm circuit 2 (*)
bit4	Overload sanitary water heaters	bit4	
bit5	Overload sanitary water pump	bit5	
bit6	Enable sanitary water second set point	bit6	
bit7	Supply voltage sequence phase failure	bit7	

READING/WRTING MACHINE STATUS			
Register 1280 / ModBus address (DEC) 1281			
MSByte	STATUS MODIFICATION	LSByte	MODIFICATION ENABLED
bit0	Device status Reading: bit=0 if device in stand-by or remote off, bit=1 if device on in chiller o heat pump Writing: allowed only device switching off setting bit=0	bit0	Enable device switch off Writing: only switch off is allowed; set bit=1 to switch off the device
bit1	Remote ON/OFF status Only reading: bit =1 device on, bit =0 device off by digital input		
bit2	Chiller status Reading: bit =1 if the device is on in chiller Writing: set bit =1 to switch on the device in chiller	bit2	Chiller status Reading: if bit =1 device on in chiller Writing: set bit=1 to switch on the device in chiller
bit3	Heat pump status Reading: bit =1 if the device is on in Heat pump Writing: set bit =1 to switch on the device in Heat pump	bit3	Heat pump status Reading: if bit =1 device on in Heat pump Writing: set bit=1 to switch on the device in Heat pump

READING/WRTING SETPOINT			
REGISTER	DESCRIPTION	SIZE (word)	NOTE
Register 768 / ModBus address (DEC) 769	Chiller Set point	1	Reading / Writing
Register 771/ ModBus address (DEC) 772	Heat pump Set point	1	Reading / Writing

READING/Writing SETPOINT

Register 33410/ ModBus address (DEC) 33411	Set point Laser (if present)	1	Reading / Writing Before the reading / writing of the parameter, the Register 1795/ Modbus address (DEC) 1796 must be fixed to 1. At the end of the reading / writing set it to 0 again. The Register 1793 / Modbus address (DEC) 1794 remains always to 0.
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RELAY STATUS

MSByte	RELAY STATUS	LSByte	RELAY STATUS
Register 2048 / ModBus address (DEC) 2049			
bit0	Alarm	bit0	Alarm
bit1	Water pump of the evaporator / supply fan	bit1	Water pump of the evaporator / supply fan
bit2	Supply water pump of the evaporator / support supply fan	bit2	Supply water pump of the evaporator / support supply fan
bit3	Antifreeze heater circuit 1 (first step)	bit3	Antifreeze heater circuit 1 (first step)
bit4	Antifreeze heater circuit 2 (second step)	bit4	Antifreeze heater circuit 2 (second step)
bit5	Condenser water pump	bit5	Condenser water pump
bit6	Condenser support water pump	bit6	Condenser support water pump
bit7	Reversing valve circuit 1	bit7	Reversing valve circuit 1
Register 2049/ModBus address (DEC) 2050			
bit0	Reversing valve circuit 2	bit0	Reversing valve circuit 2
bit1	Condensing fan 1 of the circuit 1	bit1	Condensing fan 1 of the circuit 1
bit2	Condensing fan 2 of the circuit 1	bit2	Condensing fan 2 of the circuit 1
bit3	Condensing fan 3 of the circuit 1	bit3	Condensing fan 3 of the circuit 1
bit4	Condensing fan 4 of the circuit 1	bit4	Condensing fan 4 of the circuit 1
bit5	Condensing fan 1 of the circuit 2	bit5	Condensing fan 1 of the circuit 2
bit6	Condensing fan 2 of the circuit 2	bit6	Condensing fan 2 of the circuit 2
bit7	Condensing fan 3 of the circuit 2	bit7	Condensing fan 3 of the circuit 2
Register 2050 / ModBus address (DEC) 2051			
bit0	Condensing fan 4 of the circuit 2	bit0	Condensing fan 4 of the circuit 2
bit1	Pump_down valve circuit 1	bit1	Pump_down valve circuit 1
bit2	Pump_down valve circuit 2	bit2	Pump_down valve circuit 2
bit3	Recovery valve circuit 1	bit3	Recovery valve circuit 1
bit4	Recovery valve circuit 2	bit4	Recovery valve circuit 2
bit5	Free cooling valve	bit5	Free cooling valve
bit6	Auxiliary output 1	bit6	Auxiliary output 1
bit7	Auxiliary output 2	bit7	Auxiliary output 2
Register 2051/ModBus address (DEC) 2052			
bit0	Intermittent valve compressor 1	bit0	Intermittent valve compressor 1
bit1	Intermittent valve compressor 2	bit1	Intermittent valve compressor 2
bit2	Valve for liquid injection compressor 1	bit2	Valve for liquid injection compressor 1
bit3	Valve for liquid injection compressor 2	bit3	Valve for liquid injection compressor 2
bit4	Valve 1 for sanitary water	bit4	Valve 1 for sanitary water
bit5	Valve 2 for sanitary water	bit5	Valve 2 for sanitary water
bit6	Heater 1 for sanitary water	bit6	Heater 1 for sanitary water
bit7	Heater 2 for sanitary water	bit7	Heater 2 for sanitary water
Register 2052/ModBus address (DEC) 2053			
bit0	Heater 3 for sanitary water	bit0	Heater 3 for sanitary water
bit1	Solar panel water pump	bit1	Solar panel water pump
bit2	Solar panel valve	bit2	Solar panel valve
bit3	Sanitary water pump	bit3	Sanitary water pump
bit4	Hybrid exchanger 1 circuit 1	bit4	Hybrid exchanger 1 circuit 1
bit5	Hybrid exchanger 2 circuit 1	bit5	Hybrid exchanger 2 circuit 1
bit6	Hybrid exchanger 1 circuit 2	bit6	Hybrid exchanger 1 circuit 2
bit7	Hybrid exchanger 2 circuit 2	bit7	Hybrid exchanger 2 circuit 2
Register 2053/ModBus address (DEC) 2054			
bit0	Cooling/heating circuit 1	bit0	Cooling/heating circuit 1
bit1	Cooling/heating circuit 2	bit1	Cooling/heating circuit 2

RELAY STATUS			
MSByte	RELAY STATUS	LSByte	RELAY STATUS
bit2	Defrost circuit 1	bit2	Defrost circuit 1
bit3	Defrost circuit 2	bit3	Defrost circuit 2
bit4	Circuit 1 on power request (cooling or heating)	bit4	Circuit 1 on power request (cooling or heating)
bit5	Circuit 2 on power request (cooling or heating)	bit5	Circuit 2 on power request (cooling or heating)
bit6	Sanitary water circuit on power request	bit6	Sanitary water circuit on power request
bit7	Stand-by or remote off	bit7	Stand-by or remote off
Register 2054/ModBus address (DEC) 2055			
bit0	Solenoid valve circuit 1	bit0	Solenoid valve circuit 1
bit1	Solenoid valve circuit 2	bit1	Solenoid valve circuit 2
bit2	Compressor 1	bit2	Compressor 1
bit3	Part-winding coil compressor 1	bit3	Part-winding coil compressor 1
bit4	Capacity step valve 1 compressor 1	bit4	Capacity step valve 1 compressor 1
bit5	Capacity step valve 2 compressor 1	bit5	Capacity step valve 2 compressor 1
bit6	Capacity step valve 3 compressor 1	bit6	Capacity step valve 3 compressor 1
bit7	By-pass valve compressor 1	bit7	By-pass valve compressor 1
Register 2055/ModBus address (DEC) 2056			
bit0	Compressor 2	bit0	Compressor 2
bit1	Part-winding coil compressor 2	bit1	Part-winding coil compressor 2
bit2	Capacity step valve 1 compressor 2	bit2	Capacity step valve 1 compressor 2
bit3	Capacity step valve 2 compressor 2	bit3	Capacity step valve 2 compressor 2
bit4	Capacity step valve 3 compressor 2	bit4	Capacity step valve 3 compressor 2
bit5	By-pass valve compressor 2	bit5	By-pass valve compressor 2
bit6	Compressor 3	bit6	Compressor 3
bit7	Part-winding coil compressor 3	bit7	Part-winding coil compressor 3
Register 2056/ModBus address (DEC) 2057			
bit0	Capacity step valve 1 compressor 3	bit0	Capacity step valve 1 compressor 3
bit1	Capacity step valve 2 compressor 3	bit1	Capacity step valve 2 compressor 3
bit2	Capacity step valve 3 compressor 3	bit2	Capacity step valve 3 compressor 3
bit3	By-pass valve compressor 3	bit3	By-pass valve compressor 3
bit4	Compressor 4	bit4	Compressor 4
bit5	Part-winding coil compressor 4	bit5	Part-winding coil compressor 4
bit6	Capacity step valve 1 compressor 4	bit6	Capacity step valve 1 compressor 4
bit7	Capacity step valve 2 compressor 4	bit7	Capacity step valve 2 compressor 4
Register 2057/ModBus address (DEC) 2058			
bit0	Capacity step valve 3 compressor 4	bit0	Capacity step valve 3 compressor 4
bit1	By-pass valve compressor 4	bit1	By-pass valve compressor 4
bit2	Laser activation (*)	bit2	Laser activation (*)
bit3		bit3	
bit4		bit4	
bit5		bit5	
bit6		bit6	
bit7		bit7	

ANALOGUE OUTPUTS STATUS (reading)				
Register	ModBus address (DEC)	DESCRIPTION	SIZE (word)	NOTE
2304	2305	Analogue output 1	1	0 -100 %
2305	2306	Analogue output 2	1	0 -100 %
2306	2307	Analogue output 3	1	0 -100 %
2307	2308	Analogue output 4	1	0 -100 %
2308	2309	I/O Expansion Analogue output 1	1	0 -100 %
2309	2310	I/O Expansion Analogue output 2	1	0 -100 %
2310	2311	I/O Expansion Analogue output 3	1	0 -100 %

ALARM STATUS (reading)			
MSB	DESCRIPTION	LSB	DESCRIPTION
Register 3328/ModBus address (DEC) 3329			
bit0	not valid timetable	bit0	Not used
bit1	probe 1 alarm	bit1	Not used
bit2	probe 2 alarm	bit2	Not used
bit3	probe 3 alarm	bit3	Remote keyboard 1 probe alarm
bit4	probe 4 alarm	bit4	Remote keyboard 2 probe alarm
bit5	probe 5 alarm	bit5	Compressor 1 maintenance warning
bit6	probe 6 alarm	bit6	Compressor 2 maintenance warning
bit7	Not used	bit7	Compressor 3 maintenance warning
Register 3329/ModBus address (DEC) 3330			
bit0	Compressor 4 maintenance warning	bit0	Solar panel water pump maintenance warning
bit1	Not used	bit1	Free cooling water pump maintenance warning
bit2	Not used	bit2	Configuration alarm 2
bit3	Water pump / supply fan maintenance warning	bit3	Configuration alarm 3
bit4	Evaporator 2 pump maintenance warning	bit4	Configuration alarm 4
bit5	Condenser 1 pump maintenance warning	bit5	Configuration alarm 5
bit6	Condenser 2 pump maintenance warning	bit6	Configuration alarm 6
bit7	Sanitary water pump maintenance warning	bit7	Configuration alarm 7
Register 3330/ModBus address (DEC) 3331			
bit0	Configuration alarm 8	bit0	1st remote keyboard communication failure
bit1	Configuration alarm 9	bit1	2nd remote keyboard communication failure
bit2	Configuration alarm 10	bit2	Free cooling configuration alarm
bit3	Configuration alarm 11	bit3	Sanitary water configuration alarm
bit4	Defrost alarm circuit 1	bit4	Power supply phase sequence alarm
bit5	Defrost alarm circuit 2	bit5	Configuration 1 alarm
bit6	Not used	bit6	Configuration 14 alarm
bit7	Low inlet air temperature	bit7	I/O expansion board communication alarm
Register 3331/ModBus address (DEC) 3332			
bit0	Expansion valve communication alarm	bit0	I/O expansion board probe 8 error
bit1	I/O expansion board probe 1 error	bit1	Expansion valve probe 1 error
bit2	I/O expansion board probe 2 error	bit2	Expansion valve probe 2 error
bit3	I/O expansion board probe 3 error	bit3	Expansion valve probe 3 error
bit4	I/O expansion board probe 4 error	bit4	Expansion valve probe 4 error
bit5	I/O expansion board probe 5 error	bit5	Expansion valve alarm circuit 1 (*)
bit6	I/O expansion board probe 6 error	bit6	Expansion valve alarm circuit 2 (*)
bit7	I/O expansion board probe 7 error	bit7	Not used
Register 3332/ModBus address (DEC) 3333			
bit0	Laser unit configuration alarm (AC15) (*)	bit0	Laser unit configuration alarm (AC15) (*)
bit1	Not used	bit1	Not used
bit2	Not used	bit2	Not used
bit3	Not used	bit3	Not used
bit4	Not used	bit4	Not used
bit5	Not used	bit5	Not used
bit6	Not used	bit6	Not used
bit7	Not used	bit7	Not used
Register 3333/ModBus address (DEC) 3334			
bit0	EEPROM alarm	bit0	Compressor 1 overload
bit1	Faulty clock	bit1	Compressor 2 overload
bit2	Compressor 1 high pressure	bit2	Compressor 3 overload

ALARM STATUS (reading)			
MSB	DESCRIPTION	LSB	DESCRIPTION
bit3	Compressor 2 high pressure	bit3	Compressor 4 overload
bit4	Compressor 3 high pressure	bit4	Not used
bit5	Compressor 4 high pressure	bit5	Not used
bit6	Not used	bit6	Condensing fan overload circuit 1
bit7	Not used	bit7	Condensing fan overload circuit 2
Register 3334/ModBus address (DEC) 3335			
bit0	Evaporator 1 pump overload	bit0	
bit1	Evaporator 2 pump overload	bit1	
bit2	Condenser 1 pump overload	bit2	
bit3	Condenser 2 pump overload	bit3	
bit4	Supply fan overload	bit4	
bit5	Contemporary heating and coling request (condensing unit)	bit5	
bit6		bit6	
bit7		bit7	
Register 3337/ModBus address (DEC) 3338			
bit0	Compressor 1 oil level	bit0	Condenser flow switch
bit1	Compressor 2 oil level	bit1	Low pressure switch circuit 1 (digital input)
bit2	Compressor 3 oil level	bit2	Low pressure switch circuit 2 (digital input)
bit3	Compressor 4 oil level	bit3	Low pressure switch circuit 1 (probe)
bit4	Not used	bit4	Low pressure switch circuit 2 (probe)
bit5	Not used	bit5	Pump-down circuit 1 alarm
bit6	High inlet temperature	bit6	Pump-down circuit 2 alarm
bit7	Evaporator flow switch	bit7	Pump-down at stop circuit 1 alarm
Register 3338/ModBus address (DEC) 3339			
bit0	Pump-down at stop circuit 2 alarm	bit0	High temperature compressor 4 discharge
bit1	Chiller antifreeze heater circuit 1	bit1	Not used
bit2	Chiller antifreeze heater circuit 2	bit2	Not used
bit3	Heat Pump antifreeze heater circuit 1 alarm	bit3	General alarm 1
bit4	Heat Pump antifreeze heater circuit 2 alarm	bit4	General alarm 2
bit5	High temperature compressor 1 discharge	bit5	High pressure switch circuit 1 (digital input)
bit6	High temperature compressor 2 discharge	bit6	High pressure switch circuit 2 (digital input)
bit7	High temperature compressor 3 discharge	bit7	High pressure switch circuit 1 (probe)
Register 3339/ModBus address (DEC) 3340			
bit0	High pressure switch circuit 2 (probe)	bit0	
bit1	Sanitary water flow switch	bit1	
bit3	Solar panel flow switch	bit3	
bit4	Overload sanitary water heaters	bit4	
bit5	Overload sanitary water pump	bit5	
bit6	Compressor 1 oil differential pressure switch	bit6	
bit7	Compressor 2 oil differential pressure switch	bit7	

WORKING HOURS (reading - writing)				
Register	ModBus address (DEC)	DESCRIPTION	SIZE (word)	NOTE
3584	3585	Compressor 1 running hours	1	
3585	3586	Compressor 2 running hours	1	
3586	3587	Compressor 3 running hours	1	
3587	3588	Compressor 4 running hours	1	

WORKING HOURS (reading - writing)				
3588	3589		1	
3589	3590		1	
3590	3591	Evaporator water pump 1 or Supply fan running hours	1	
3591	3592	Evaporator water pump 2 or Supply fan running hours	1	
3592	3593	Condenser pump 1 running hours	1	
3593	3594	Condenser pump 2 running hours	1	
3594	3595	Sanitary water pump running hours	1	
3595	3596	Solar panel running hours	1	
3596	3597	Free cooling water pump running hours	1	

NUMBER OF ACTIVE ALARMS PER HOUR				
NUMBER OF ACTIVE ALARMS PER HOUR (reading)				
Register 3840	ModBus address (DEC) 3841	COMPRESSOR 1 OIL LEVEL		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3841	ModBus address (DEC) 3842	COMPRESSOR 2 OIL LEVEL		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3842	ModBus address (DEC) 3843	COMPRESSOR 3 OIL LEVEL		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3843	ModBus address (DEC) 3844	COMPRESSOR 4 OIL LEVEL		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3844	ModBus address (DEC) 3845			
MSByte		N° of activation	LSByte	Alarm threshold
Register 3845	ModBus address (DEC) 3846			
MSByte		N° of activation	LSByte	Alarm threshold
Register 3846	ModBus address (DEC) 3847	HIGH TEMPERATURE (CONFIGURABLE PROBE)		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3847	ModBus address (DEC) 3848	NOT USED		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3848	ModBus address (DEC) 3849	NOT USED		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3849	ModBus address (DEC) 3850	LOW TEMPERATURE /PRESSURE CIRCUIT 1 (Digital input)		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3850	ModBus address (DEC) 3851	LOW TEMPERATURE /PRESSURE CIRCUIT 2 (Digital input)		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3851	ModBus address (DEC) 3852	LOW TEMPERATURE /PRESSURE CIRCUIT 1 (Probe)		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3852	ModBus address (DEC) 3853	LOW TEMPERATURE /PRESSURE CIRCUIT 2 (Probe)		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3853	ModBus address (DEC) 3854	PUMP_DOWN AT THE START-UP CIRCUIT 1		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3854	ModBus address (DEC) 3855	PUMP_DOWN AT THE START-UP CIRCUIT 2		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3855	ModBus address (DEC) 3856	PUMP_DOWN IN SWITCHING OFF CIRCUIT 1		
MSByte		N° of activation	LSByte	Alarm threshold

Register 3856	ModBus address (DEC) 3857	PUMP_DOWN IN SWITCHING OFF CIRCUIT 2		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3857	ModBus address (DEC) 3858	ANTIFREEZE CHILLER CIRCUIT 1		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3858	ModBus address (DEC) 3859	ANTIFREEZE CHILLER CIRCUIT 2		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3859	ModBus address (DEC) 3860	ANTIFREEZE HEAT PUMP CIRCUIT 1		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3860	ModBus address (DEC) 3861	ANTIFREEZE HEAT PUMP CIRCUIT 2		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3861	ModBus address (DEC) 3862	COMPRESSOR 1 DISCHARGE TEMPERATURE		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3862	ModBus address (DEC) 3863	COMPRESSOR 2 DISCHARGE TEMPERATURE		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3863	ModBus address (DEC) 3864	COMPRESSOR 3 DISCHARGE TEMPERATURE		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3864	ModBus address (DEC) 3865	COMPRESSOR 4 DISCHARGE TEMPERATURE		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3865	ModBus address (DEC) 3866			
MSByte		N° of activation	LSByte	Alarm threshold
Register 3866	ModBus address (DEC) 3867			
MSByte		N° of activation	LSByte	Alarm threshold
Register 3867	ModBus address (DEC) 3868	GENERAL ALARM 1		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3868	ModBus address (DEC) 3869	GENERAL ALARM 2		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3869	ModBus address (DEC) 3870	CIRCUIT 1 HIGH PRESSURE ALARM (DIGITAL INPUT)		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3870	ModBus address (DEC) 3871	CIRCUIT 2 HIGH PRESSURE ALARM (DIGITAL INPUT)		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3871	ModBus address (DEC) 3872	CIRCUIT 1 HIGH PRESSURE ALARM (PROBE)		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3872	ModBus address (DEC) 3873	CIRCUIT 2 HIGH PRESSURE ALARM (PROBE)		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3873	ModBus address (DEC) 3874	NOT USED		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3874	ModBus address (DEC) 3875	NOT USED		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3875	ModBus address (DEC) 3876	OVERLOAD SANITARY WATER HEATERS		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3876	ModBus address (DEC) 3877	OVERLOAD SANITARY WATER PUMP		
MSByte		N° of activation	LSByte	Alarm threshold

Register 3877	ModBus address (DEC) 3878	PRESSURE OIL DIFFERENTIAL SWITCH CIRCUIT 1 OR COMPRESSOR 1		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3878	ModBus address (DEC) 3879	PRESSURE OIL DIFFERENTIAL SWITCH CIRCUIT 1 OR COMPRESSOR 2		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3879	ModBus address (DEC) 3880	NOT USED		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3880	ModBus address (DEC) 3881	OVERLOAD COMPRESSOR 1		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3881	ModBus address (DEC) 3882	OVERLOAD COMPRESSOR 2		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3882	ModBus address (DEC) 3883	OVERLOAD COMPRESSOR 3		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3883	ModBus address (DEC) 3884	OVERLOAD COMPRESSOR 4		
MSByte		N° of activation	LSByte	Alarm threshold
Register 3884	ModBus address (DEC) 3885			
MSByte		N° of activation	LSByte	Alarm threshold
Register 3885	ModBus address (DEC) 3886			
MSByte		N° of activation	LSByte	Alarm threshold

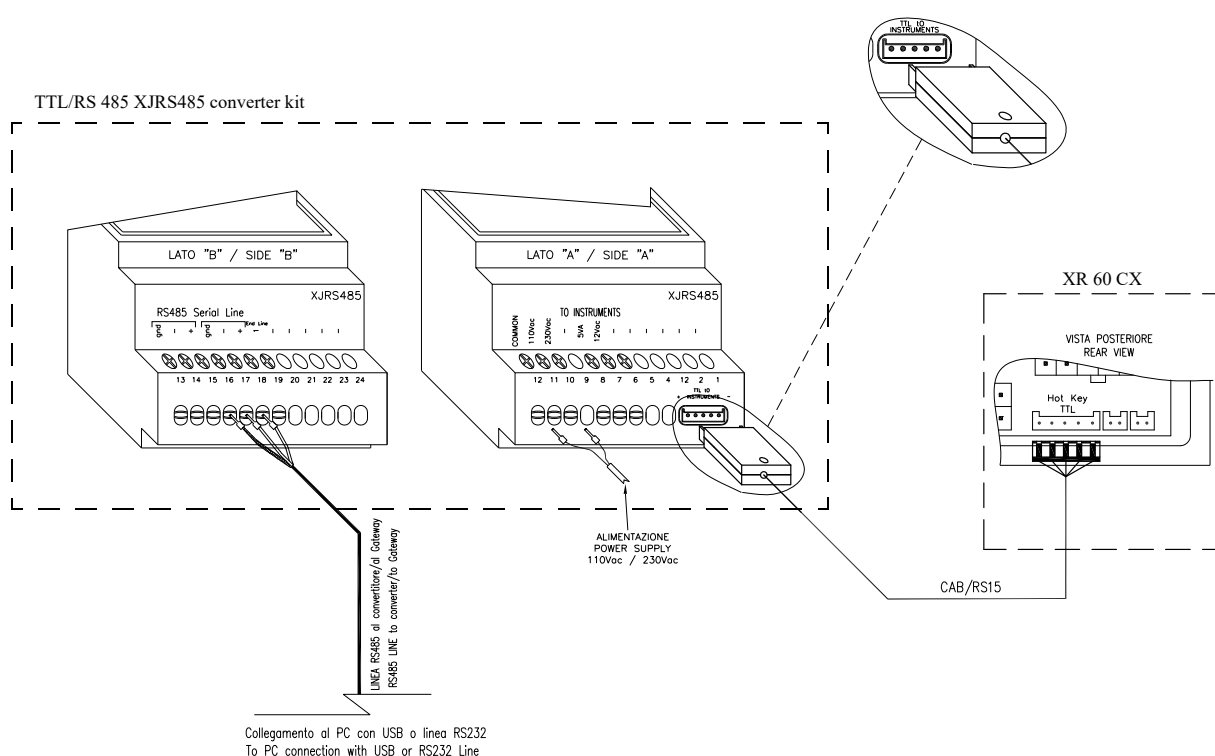
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8. MODBUS-RTU

8.1 Connection drawing

The drawing below shows how to connect the unit to a Modbus network:



8.2 Configuration

Communication is configured as follows:

Baud Rate = 9600 bps

Data Length = 8 bit

Parity = None

Stop Bit = 1

8.3 How to set the device address

Press **SET** +  buttons for 3 sec. to enter the programming procedure, then press **SET** until reaching **Adr** parameter.

Use  and  buttons to modify the value.

Press **SET** again to store the new value.

8.4 Variables

PROBE STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
257	Regulation probe value	1	
258	1st defrost probe value	1	
259	2nd defrost probe value	1	
260	Fan temperature probe value	1	
261	Cold store temperature alarm probe value	1	
262	Condenser temperature alarm probe value	1	
263	Auxiliary regulation probe value	1	
264	Dynamic setpoint probe value	1	
265	P1 probe value	1	
266	P2 probe value	1	
267	P3 probe value	1	
268	P4 probe value	1	

NOTE

Consult to the electrical diagram of the machine to see that the probe is related to the "value of probe 1." PB1= Probe value 1 / PB2= Probe value 2 /

DIGITAL INPUT STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
513	On/Off	1	
520	Digital input	1	
521	P1 probe alarm	1	
522	P2 probe alarm	1	
523	P3 probe alarm	1	
524	P4 probe alarm	1	
525	Cold store high temperature alarm	1	
526	Cold store low temperature alarm	1	
527	Condenser high temperature alarm	1	
528	Condenser low temperature alarm	1	
529	Unspecified alarm from digital input: EAL	1	
530	Unspecified alarm from digital input: BAL	1	
531	Door open alarm	1	
532	EEPROM alarm	1	
533	Temperature unit of measurement: °C - °F	1	
534	Temperature value resolution: decimal - integer	1	
535	ON/OFF relay output	1	
536	1st defrost relay output	1	
537	2nd defrost relay output	1	
538	Alarm relay output	1	
540	Fan relay output	1	
542	Neutral zone relay output	1	
543	1st compressor relay output	1	
544	2nd compressor relay output	1	
545	Buzzer output	1	
548	Draining heating element relay output	1	
549	Condenser relay output	1	

FUNCTION PARAMETERS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
1537	Regulation setpoint	1	
1540	Setpoint 2 (if enabled)	1	

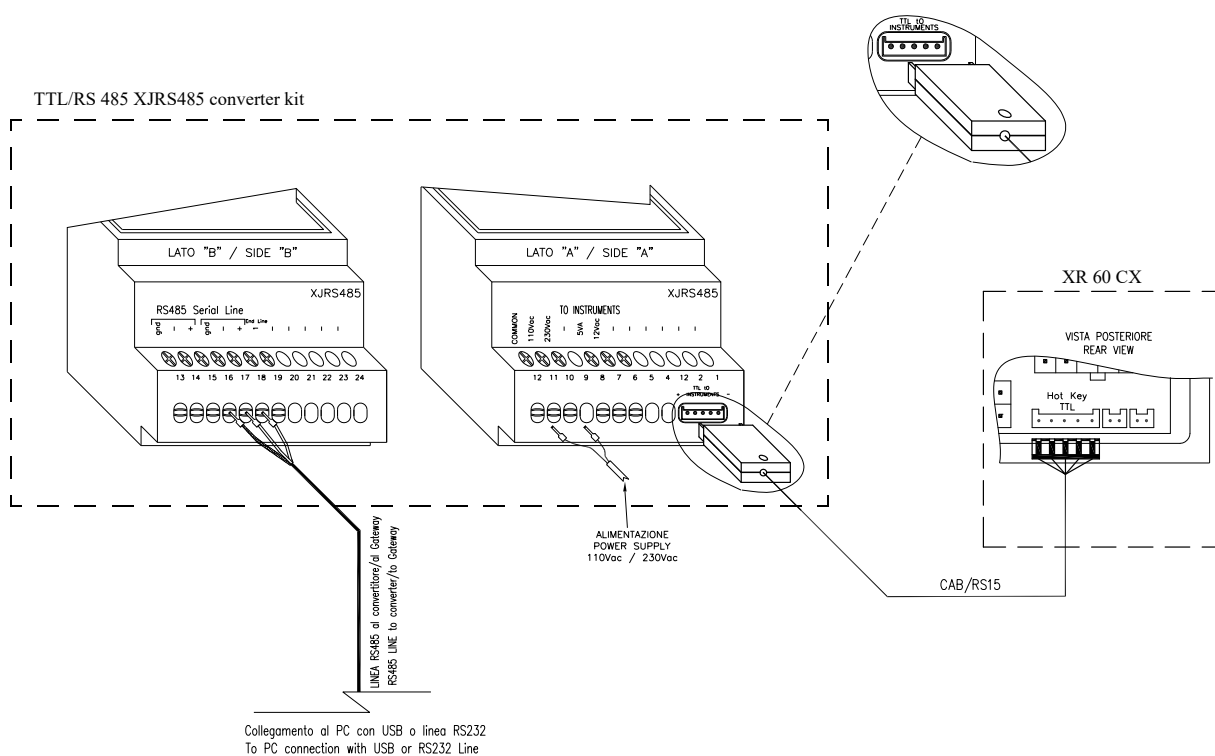
XR60CX DE X - DE H



9. MODBUS-RTU

9.1 Connection drawing

The drawing below shows how to connect the unit to a Modbus network:



9.2 Configuration

Communication is configured as follows:

Baud Rate = 9600 bps

Data Length = 8 bit

Parity = None

Stop Bit = 1

9.3 How to set the device address

Press **SET** +  buttons for 3 sec. to enter the programming procedure, then press **SET** until reaching **Adr** parameter.

Use  and  buttons to modify the value.

Press **SET** again to store the new value.

9.4 Variables

PROBE STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
265	P1 probe value	1	
266	P2 probe value	1	
268	P4 probe value	1	

NOTE

Consult to the electrical diagram of the machine to see that the probe is related to the "value of probe 1." PB1= Probe value 1 / PB2= Probe value 2 /

DIGITAL INPUT STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE bit	NOTE
513	On/Off	1	Reading/writing
520	Digital input	1	
521	P1 probe alarm	1	
522	P2 probe alarm	1	
524	P4 probe alarm	1	
525	Pb1 high temperature alarm	1	
526	Pb1 low temperature alarm	1	
527	Pb2 high temperature alarm	1	
528	Pb2 low temperature alarm	1	
529	Unspecified alarm from digital input: EAL	1	
530	Unspecified alarm from digital input: BAL	1	
531	Compressor maintenance alarm	1	
532	EEPROM alarm	1	
548	Pressure switch alarm from digital input PAL	1	

DIGITAL OUTPUT STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE bit	NOTE
536	Condensate drain output relay	1	
543	Compressor output	1	

USER PARAMETERS (reading-writing)			
MODBUS ADDRESS (DEC)	PARAMETER	DISPLAY REF.	SIZE (Word 16bit)
769	Summer setpoint	SEE	1
770	Winter setpoint (if available)	SEI	1

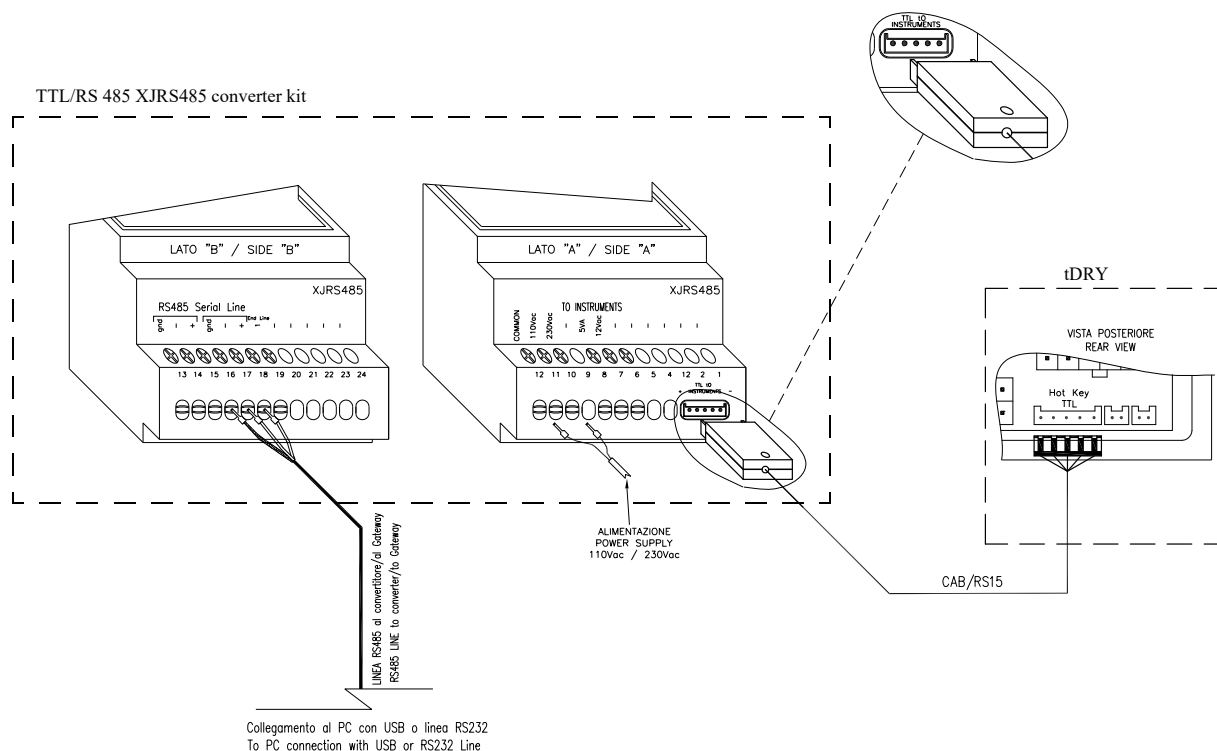
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10. MODBUS-RTU

10.1 Connection drawing

The drawing below shows how to connect the unit to a Modbus network:



10.2 Configuration

Communication is configured as follows:

Baud Rate = 9600 bps

Data Length = 8 bit

Parity = None

Stop Bit = 1

10.3 How to set the device address

Press **SET** +  buttons for 3 sec. to enter the programming procedure, then press **SET** until reaching **CF30** parameter.

Use  and  buttons to modify the value.

Press **SET** again to store the new value.

10.4 Variables

IDENTIFICATION AND CONFIGURATION (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
1	Product family code/Firmware version	1	Bit15-Bit8: product family code; Bit7-Bit0: Firmware Version
6	Firmware version date	1	Bit15-Bit11: Day; Bit10-Bit7: Month; Bit6-Bit0: Year

PROBE STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
257	1st probe value	1	
259	2nd probe value	1	
261	3rd probe value	1	
263	4th probe value	1	

NOTE

Consult to the electrical diagram of the machine to see that the probe is related to the "value of probe 1." PB1= Probe value 1 / PB2= Probe value 2 /

DIGITAL INPUT STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
513	Digital input status (1st)	1	Bit0 Condensing fan thermal overload Bit1 Remote off Bit2 Unspecified user alarm Bit3 Energy saving Bit4 Anti-freeze alarm Bit5 Bit6 Bit7 Bit8 Bit9 Compressor 1 thermal overload Bit10 Compressor 2 thermal overload Bit11 Compressor 1 and 2 global thermal overload Bit12 Compressor discharge high temperature Bit13 Low pressure switch Bit14 High pressure switch Bit15 Condensate drain alarm

USER PARAMETERS (write)			
MODBUS ADDRESS (DEC)	PARAMETER	RIF. DISPLAY	SIZE (Word 16bit)
769	Summer setpoint (during writing the display must show the main mask)	St01	1

UNIT FUNCTIONAL STATUS (read-write)

MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
1281	Unit status (1st)	1	Bit0 Unit on-off (Send Modbus command with only Bit0 on 1 to switch the unit off - during writing the display must show the main mask) Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Bit7 Bit8 Unit on/off0: stand-by or remote-off 1: on (read only) Bit9 Bit10 Bit11 Bit12 Bit13 Bit14 Bit15
1283	Unit status (3rd)	1	Bit0 Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Switch-on enabling (Send Modbus command with only Bit6 and Bit14 on 1 to switch the unit on - during writing the display must show the main mask) Bit7 Bit8 Bit9 Bit10 Unit on/off from digital input (reading) Bit11 Bit12 Bit13 Bit14 Switch-on (Send Modbus command with only Bit6 and Bit14 on 1 to switch the unit on - during writing the display must show the main mask) Bit15

DIGITAL OUTPUT STATUS (reading)

MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
2049	Relay output status (1st)	1	Bit0 Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Bit7 Bit8 Compressor 1 Bit9 Compressor 2 Bit10 Condensate drain Bit11 Step 1 condensing fans Bit12 Step 2 condensing fans Bit13 Alarms Bit14 Anti-freeze heaters Bit15

ANALOGUE OUTPUT STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
2305	Analogue output 1 value	1	
2307	Analogue output 2 value	1	

ALARM STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
3332	Functional alarms (4th)	1	Bit0 A24 - Evaporator 1 outlet air temperature Bit1 A25 - Evaporator 2 outlet air temperature Bit2 C1Mn - Compressor 1 maintenance Bit3 C2Mn - Compressor 2 maintenance Bit4 Bit5 Bit6 Bit7 Bit8 P1 - Pb1 probe damaged Bit9 P2 - Pb2 probe damaged Bit10 P3 - Pb3 probe damaged Bit11 P4 - Pb4 probe damaged Bit12 A05 - Condensate drain Bit13 A21 - High inlet air temperature Bit14 A22 - High outlet air temperature Bit15 A23 - Evaporator outlet air temperature
3333	Functional alarms (5th)	1	Bit0 A06: Compressor 1 thermal overload Bit1 A07: Compressor 2 thermal overload Bit2 A08: Overall thermal overload Bit3 A10: Unspecified user Bit4 A11: inlet air low temperature alert Bit5 A12: outlet air low temperature alert Bit6 A13: Evaporator outlet air low temperature alert Bit7 A14: Evaporator 1 outlet air low temperature alert Bit8 Bit9 AFr: Network frequency Bit10 A01: High pressure switch Bit11 A03: Condensing HT/HP from probe Bit12 A09: Compressor discharge high temperature Bit13 Bit14 A02: Low pressure switch Bit15 A04: LT/LT from probe
3334	Functional alarms (6th)	1	Bit0 Bit1 Bit2 Bit3 Bit4 Bit5 Bit6 Bit7 Bit8 A15: Evaporator 2 outlet air low temperature alert Bit9 A16: Low inlet air temperature Bit10 A17: Low outlet air temperature Bit11 A18: Evaporator low outlet air temperature Bit12 A19: Evaporator 1 low outlet air temperature Bit13 A20: Evaporator 2 low outlet air temperature Bit14 AtC: Condensing fan thermal overload Bit15

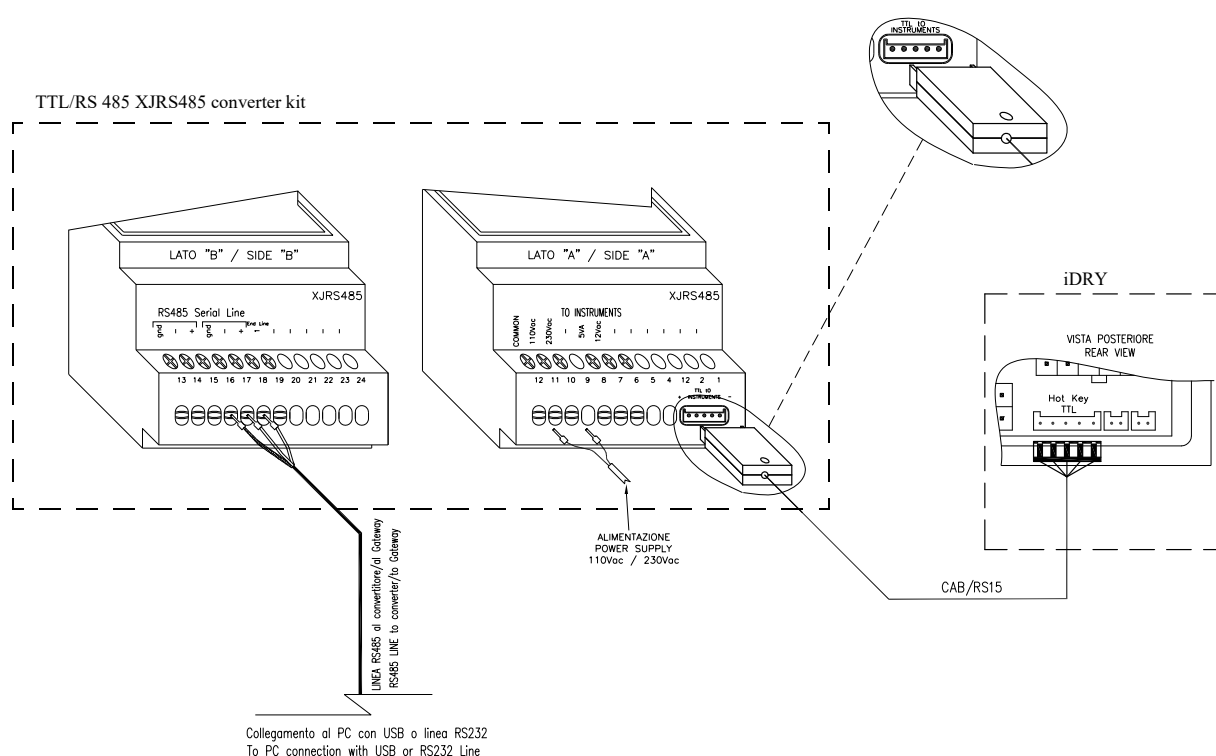
LOAD HOURS (read-write)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
3585	Compressor 1 operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	
3586	Compressor 2 operating hours (Set to 0 to reset -during writing the main mask must be displayed)	1	

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11. MODBUS-RTU

11.1 Connection drawing

The drawing below shows how to connect the unit to a Modbus network:



11.2 Configuration

Communication is configured as follows:

Baud Rate = 9600 bps

Data Length = 8 bit

Parity = None

Stop Bit = 1

11.3 How to set the device address

Press **SET** +  buttons for 3 sec. to enter the programming procedure, then press **SET** until reaching **Adr** parameter.

Use  and  buttons to modify the value.

Press **SET** again to store the new value.

11.4 Variables

IDENTIFICATION AND CONFIGURATION (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
1	Product family code/Firmware version	1	Bit15-Bit8: product family code; Bit7-Bit0: Firmware Version
2		1	
3		1	
4		1	
5		1	
6	Firmware version date	1	Bit15-Bit11: Day; Bit10-Bit7: Month; Bit6-Bit0: Year
7		1	
8		1	
9		1	
10		1	
11		1	
12		1	

PROBE STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	NOTE
257	Regulation probe value	1	
258			
259			
260			
261	Cell temperature alarm probe value	1	
262	Condenser temperature alarm probe value	1	
263			
264			
265	P1 probe value	1	
266	P2 probe value	1	
267	P3 probe value	1	
268	P4 probe value	1	

NOTE

Consult to the electrical diagram of the machine to see that the probe is related to the "value of probe 1." PB1= Probe value 1 / PB2= Probe value 2 /

DIGITAL INPUT STATUS (reading)			
MODBUS ADDRESS (DEC)	PARAMETER	SIZE Bit	NOTE
513	On/Off	1	Reading/writing
514		1	
515		1	
516		1	
517		1	
518		1	
519		1	
520	Digital input	1	
521	Probe P1 alarm	1	
522	Probe P2 alarm	1	
523	Probe P3 alarm	1	
524	Probe P4 alarm	1	
525	Cell high temperature alarm	1	
526	Cell low temperature alarm	1	
527	Condenser high temperature alarm	1	
528	Condenser low temperature alarm	1	
529	General alarm from digital input: EAL	1	
530	General alarm from digital input: BAL	1	

531	Compressor maintenance alarm	1	
532	EEPROM alarm	1	
533	Temperature unit of measurement: °C - °F	1	
534	Temperature value resolution: decimal - integer	1	
535	ON/OFF relay output	1	
536	Condensate drain relay output	1	
537		1	
538	Alarm relay output	1	
539		1	
540		1	
541	Pump relay output	1	
542	Neutral zone relay output	1	
543	1st compressor relay output	1	
544		1	
545	Buzzer output	1	
546		1	
547		1	
548	Pressure switch alarm from digital input: PAL	1	

USER PARAMETERS (read-write)

MODBUS ADDRESS (DEC)	PARAMETER	DISPLAY REF.	SIZE (Word 16bit)
769	Summer setpoint	SEE	1
770	Winter setpoint	SEI	1

FUNCTION PARAMETERS (reading)

MODBUS ADDRESS (DEC)	PARAMETER	SIZE (Word 16bit)	DISPLAY REF.
1537	Regulation setpoint	1	
1538	High temperature alarm setpoint	1	
1539	Low temperature alarm setpoint		
1540			
1541	High temperature alarm setpoint 2		
1542	Low temperature alarm setpoint 2		

